

Honeywell

Signum Model 2043 Printer Operation and Service

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3/98

TotalPlant

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About This Document

Abstract

This publication supports the operation and service of the Signum 2043 printer.

Revision Notes

The following list provides notes concerning all revisions of this document.

DocID	RevID	Date	Notes
PR30100	1.0	1/97	Initial Release
PR30100A	2.0	3/98	Revised to remove the unsupported Test and Diagnostic procedures

About This Document

References

Honeywell documents

The following list identifies all Honeywell documents that may be sources of reference for the material discussed in this publication.

Document Title	Doc. ID	Binder
LCN System Installation	SW20-500	TP 3025

Contacts

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1. Signum Printer Operation

1.1 Overview

Introduction

The Signum 2043 printer is a 136 column printer that is capable of 400 characters per second with an ASPI protocol. Optionally, the printer is capable of using a color ribbon that results in seven printable colors.

Two models

Two models of the printer are available: a 120 Vac model and a 240 Vac model. Only the 240 Vac model is CE Compliant.

Illustrations

Figures 1-1 and 1-2 are front and rear views of the Signum 2043 printer, respectively. Figure 1-3 is a pictorial of the printer's components.

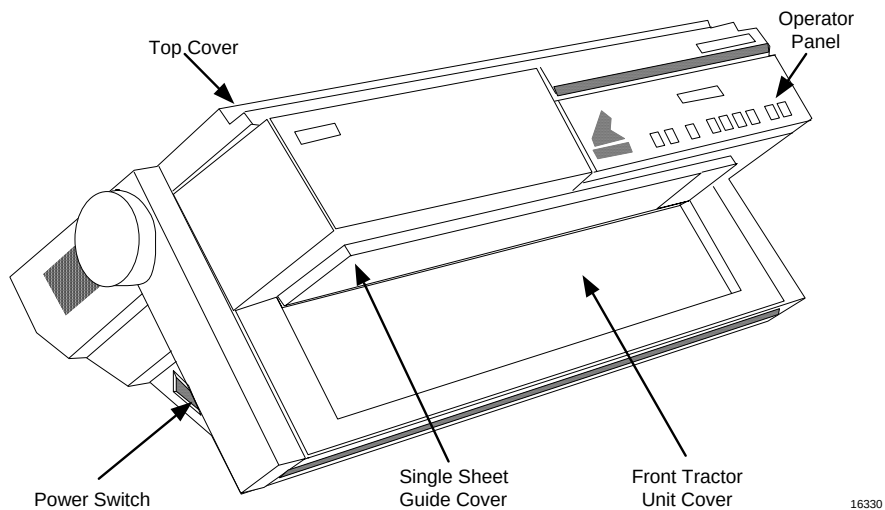


Figure 1-1 Front View of Signum 2043 Printer

Illustration

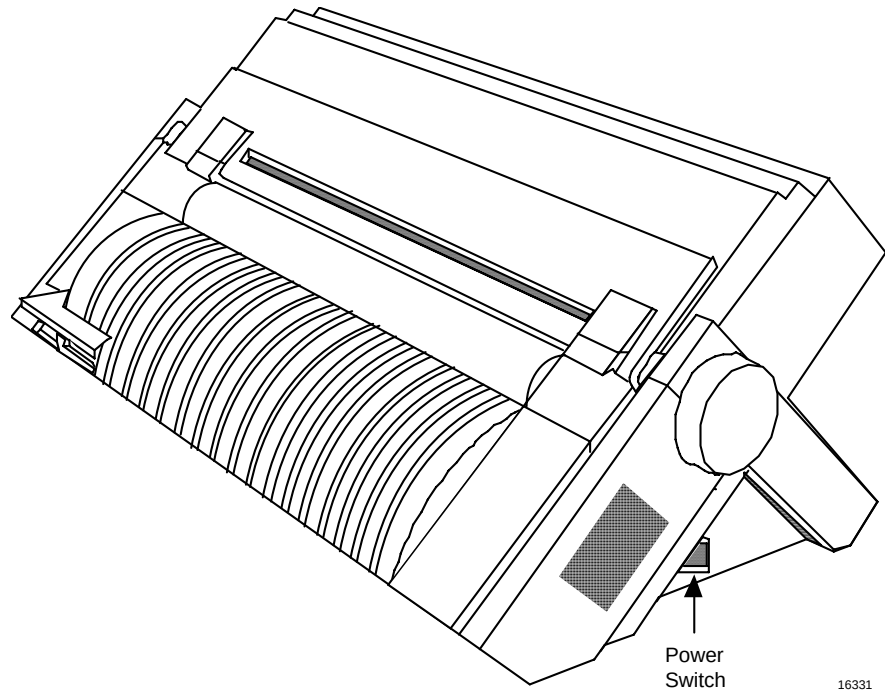


Figure 1-2 Rear View of Signum 2043 Printer

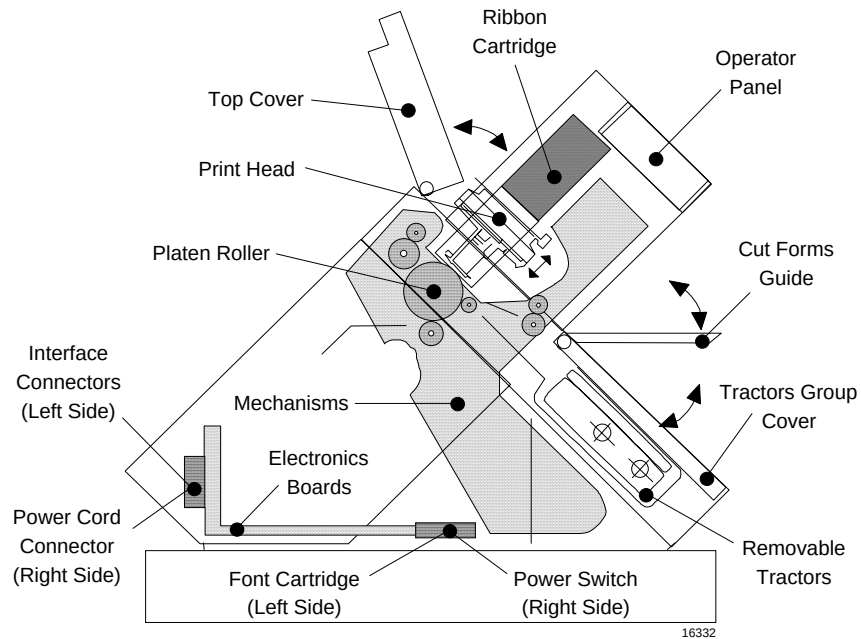


Figure 1-3 Signum 2043 Printer Pictorial

1.2 Operator Panel Description

Introduction

The printer's Operator Panel allows you to perform many of the printer functions that include paper path selection, font selection, and printer configuration setup. The Operator Panel is located at the right front side of the printer. Figure 1-4 is an illustration of the Operator Panel.

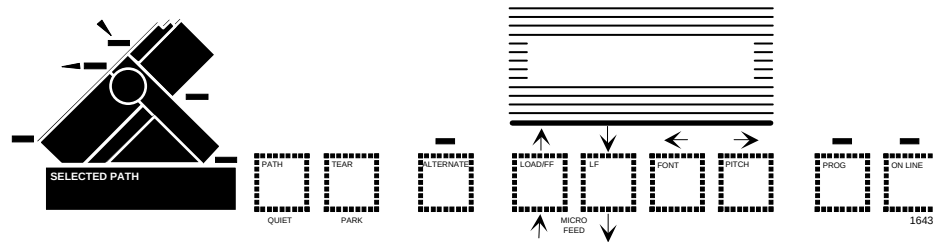


Figure 1-4 Operator Panel Layout

Description

The Operator Panel has the following features.

- A 16-character display (Liquid Crystal) that displays messages related to the printing functions and the operational state of the printer.
- Three function mode indicators indicate which function mode (**NORMAL/PROG/ALTERNATE**) is active. The indicators are green.
- Five paper path indicators that indicate which paper path is active. The indicators are green.
- Nine function keys allow you to change the operating state as necessary. They are selected by pressing the button within the squared area that represents the function. Each function key activates one of the three function modes.
 - In the Program function mode, the red symbols (↑ ↓ ← →) above the keys are valid.
 - In the Normal function mode, the legend inside the key area is valid.
 - In the Alternate function mode, the symbols (↑ ↓) and legends (QUIET, PAGE, MICRO FEED) under the key are valid.

Display messages

The display messages can be divided into three main groups.

- Printer status messages—Provide information about the current operational state of the printer.
- Printer setup messages—Messages displayed during the printer setup procedure.
- Printer error messages—Printer faults. See the *Universal Station Service* or *Universal Station Service (Ergonomic) Service* manual for printer fault or error information.

Printer status messages

Table 1-1 describes the printer status messages.

Table 1-1 Printer Status Messages

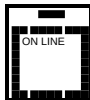
Message	Printer Status or State
xxxxxx (font) xx (cpi)	In Wait, Ready, or Print state
SELF TEST	Self Test printing state
ALTERNATE	Alternate function mode
CHANGE NVM	Change in NVM (Non Volatile Memory) Only when power is first applied or when the firmware EPROM is changed.
LOAD REAR PUSH LOAD FRONT PUSH LOAD FRONT PULL LOAD PUSH PULL LOAD MANUAL FORM LOAD ASF1 LOAD ASF2 LOAD ASF3 LOAD RR.PSH&MF LOAD FR.PSH&MF LOAD FR.PLL&MF LOAD PSH.PLL&MF	In Wait state with no paper
MANUAL FORM INSERT PAPER	Overlay condition request
QUIET PRINT OFF QUIET PRINT ON	Quiet state

Table 1-1 Printer Status Messages

Message	Printer Status or State
MICRO FEED DOWN MICRO FEED UP	Micro Feed state
EJECT PAPER PARK: SURE?	Eject/Park Paper state
RESET & BREAK	Break state

Indicators

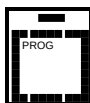
The three function mode indicators provide the following information.



The **ON LINE** indicator (green) is lit when the printer is able to receive and print data (Ready state).

The indicator will blink when data is present in the input buffer, and the printer is in the Wait state.

The **ON LINE** indicator is not lit when the printer is disabled and no data is present in the input buffer during initialization, menu, or test procedures, or in the Wait state.



The **PROG** (Program) indicator is lit when a Program function mode has been selected (Power On Configuration mode or Program Configuration mode).



The **ALTERNATE** indicator is lit when the Alternate function mode is selected.

SELECTED PATH indicators

Five indicators are used to indicate the paper path that has been selected. The indicators are identified in Figure 1-5. The five path indicators are:

- ASF Path (not used)
- Front Pull Fanfold
- Rear Push Fanfold
- Front Single Sheet
- Front Push Fanfold

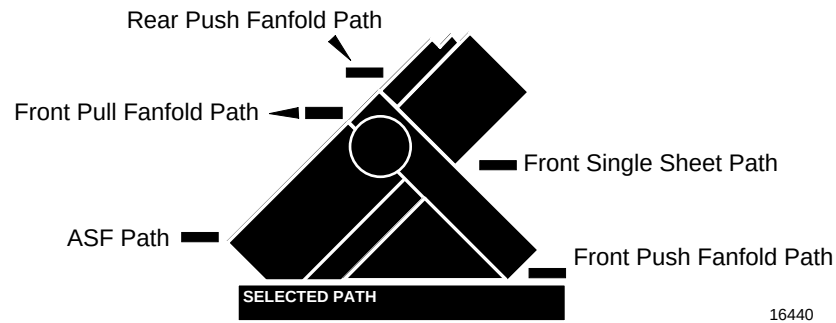


Figure 1-5 SELECTED PATH Display

Function keys

There are nine function keys. Two of the function keys are used to select two of the three function modes.

Normal function mode

The Normal function mode is the default mode (not Alternate or Program).

Program function mode

The Program function mode is selected by pressing the **PROG** key.

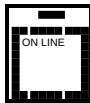
If the key is pressed while applying power to the printer, the Power On Configuration mode will be entered.

If the key is pressed while the printer is in the Ready state while not printing, or in the Wait state (the **ON LINE** indicator is not lit), the Program Configuration mode will be entered.

Alternate function mode

The Alternate function mode is selected by pressing the **ALTERNATE** key. This function key is disabled when the Program function mode is active.

ON LINE key



The **ON LINE** key is a multi-functional key that provides functions in the Normal and Program function modes. It is not functional in the Alternate function mode.

ON LINE key (Normal function mode)

Depending upon the state of the printer, pressing the **ON LINE** key provides different results.

Wait state:

- If the printer is in the Wait state without an internal error or fault, pressing the **ON LINE** key causes the printer to enter the Ready state.
- If the printer is in the Wait state because the printer is out of paper, pressing the **ON LINE** key causes data in the input buffer to be printed after the printer has been reloaded with paper.
- If the **TEAR** key was previously pressed, pressing the **ON LINE** key causes the paper to be moved backward from the tear position to the print line position.
- When a jam or printer alarm condition exists, pressing the **ON LINE** key after correcting the fault condition cancels the error message.
- When loading paper in the Front Push-Rear Pull Path mode, pressing the **ON LINE** key verifies the paper insertion.

Ready state:

- Pressing the **ON LINE** key places the printer in the Off Line state.
- If printing is in progress, pressing the **ON LINE** key stops the printing at the end of the print line.

Applying power to the printer while holding the **ON LINE** key, starts the printer test.

Pressing the **ON LINE** key during the printer test, stops the printer test.

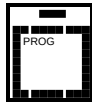
ON LINE key (Program function mode)

Pressing the **ON LINE** key clears the input buffer and sends a break (250 Msec) on the serial interface. A **RESET & BREAK** message is displayed.

ON LINE (Alternate function mode)

Not applicable.

PROG key



The **PROG** key is a multi-functional key that provides functions in the Normal and Program function modes. It is not functional in the Alternate function mode.

PROG key (Normal function mode)

If the printer is in the Wait state (the **ON LINE** indicator is not lit) or Ready state and not printing, pressing the **PROG** key causes the printer to enter the Program function mode and the red symbols above the function keys will be enabled.

If the **PROG** key is pressed while applying power to the printer, the printer will enter the Power On Configuration mode.

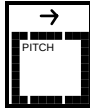
PROG key (Program function mode)

Pressing the **PROG** key twice causes the printer to exit the Program function mode. The **PROG** indicator will extinguish.

PROG key (Alternate function mode)

Not applicable.

PITCH key



The **PITCH** key is a multi-functional key that provides functions in the Normal and Program function modes. It is not functional in the Alternate function mode.

PITCH key (Normal function mode)

If the printer is in the Ready state and not printing or in the Wait state (the **ON LINE** indicator is extinguished), pressing the **PITCH** key causes the printer to enter the Pitch mode which allows selection of horizontal spacing. The various pitches will be displayed in the following sequence.

10 CPI - 12 CPI - 15 CPI - 16.7 CPI - 5 CPI - 6 CPI
7.5 CPI - 8.4 CPI

In the Print state, pressing the **PITCH** key changes the pitch at the end of the line.

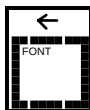
PITCH key (Program function mode)

Pressing the **LF** key allows you to scan forward to select the desired value from the available setup values.

PITCH key (Alternate function mode)

Not applicable.

FONT key



The **FONT** key is a multi-functional key that provides functions in the Normal and Program function modes. It is not functional in the Alternate function mode.

FONT key (Normal function mode)

In the Ready or Wait state, pressing the **FONT** key causes the printer to enter the Font mode to select the desired font at the selected pitch. The available fonts will be displayed in the following sequence.

Draft - Courier - OCR-B - Gothic - OCR-A

In the Print state, pressing the **FONT** key changes the pitch at the end of the line.

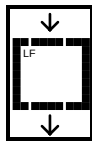
FONT key (Program function mode)

Pressing the **LF** key allows you to scan backward to select the desired value from the available setup values.

FONT key (Alternate function mode)

Not applicable.

LF key



The **LF** (Line Feed) key is a multi-functional key that provides functions in the Normal, Program, and Alternate modes.

LF key (Normal function mode)

In the Ready or Wait state, pressing the **LF** key advances the paper to the next printable line. Holding the **LF** key causes a sequence of line feeds.

In the Print state, the **LF** key has no effect.

LF key (Program function mode)

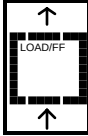
Pressing the **LF** key allows you to scan downward to select the desired value from the available setup values.

LF Key (Alternate function mode)

Pressing the **LF** key causes the paper to be moved downward in micro steps at increasing speed. A **MICRO FEED DOWN** message will be displayed. Holding the key causes a continuous movement.

To exit the Micro Feed mode, press the **ALTERNATE** key.

LOAD/FF key



The **LOAD/FF** (Load/Form Feed) key is a multi-functional key that provides functions in the Normal, Program, and Alternate function modes.

LOAD/FF key (Normal function mode)

In the Ready or Wait state and with paper in the printer's paper path, pressing the **LOAD/FF** key clears the paper path by executing a form feed. The form feed moves the paper up to the first physical line of the next form to align the perforation with the tear bar.

In the Ready or Wait state and with no paper in the printer's paper path, pressing the **LOAD/FF** key loads the paper according to the type of paper selected in the printer's setup.

LOAD/FF key (Program function mode)

Pressing the **LOAD/FF** key allows you to scan backward to select the desired value from the available setup values.

LOAD/FF key (Alternate function mode)

Pressing the **LOAD/FF** key causes the paper to be moved in micro steps at increasing speed. A MICRO FEED UP message will be displayed. Holding the key causes a continuous movement of the paper. To cancel the function, press the **ALTERNATE** key.

ALTERNATE key



The **ALTERNATE** key is a multi-functional key that provides functions in the Normal and Alternate function modes. It is not functional in the Program function mode.

ALTERNATE key (Normal function mode)

In the Ready or Print (only when the Quiet mode is active) state, pressing the **ALTERNATE** key causes the Alternate function to be entered and the Alternate indicator to illuminate. An **ALTERNATE** message will be displayed.

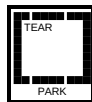
ALTERNATE key (Program function mode)

Not applicable.

ALTERNATE (Alternate function mode)

Pressing the **ALTERNATE** key cancels the Alternate function mode and returns the printer to the Normal function mode. The **ALTERNATE** indicator extinguishes.

TEAR key



The **TEAR** key is a multi-functional key that provides functions in the Normal and Alternate function modes. It is not functional in the Program function mode.

TEAR key (Normal function mode)

In the Ready or Wait state, pressing the **TEAR** key moves fanfold paper up to the tear line position. If the paper is not torn off within 10 seconds, the paper will move back to its initial position.

In the Ready or Wait state with the Automatic Tear mode enabled, pressing the **TEAR** key overrides the Automatic Tear mode.

In the Print state, pressing the **TEAR** key has no effect.

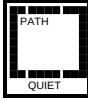
TEAR key (Program function mode) — Not applicable.

TEAR (Alternate function mode)

Pressing the **TEAR** key places the printer in the Paper Park mode. A **PARK: SURE?** message is displayed. After confirmation, pressing the **TEAR** key causes the printer to park the paper fanfold that is currently in front of the print head. If the paper sensor detects that paper is still in front of the print head, the paper movement will be stopped because it is assumed that there is a paper jam. To clear the paper jam, press the **ON LINE** key.

To exit the Park mode, press the **TEAR** key a second time.

PATH key



The **PATH** key is a multi-functional key that provides functions in the Normal and Alternate function modes. It is not functional in the Program function mode.

PATH key (Normal function mode)

If the printer is not loaded with paper in the Wait state, it is possible to select one of the paper paths that are available during the Power On Configuration mode setup.

Pressing the **PATH** key selects another paper path from the configurable paper paths.

If the **LOAD** key is pressed within 2 seconds after selecting a paper path, the paper is loaded into the newly selected paper path. Otherwise, the paper is loaded into the original paper path.

If the printer is loaded with paper in the Wait state, pressing the **PATH** key for at least 1/2 second parks the fanfold paper. The printer enters the Paper Empty mode. The **PATH** key can then be used to select a new paper path from the paths that were setup in Power On Configuration.

If the printer is in the Ready or Print state, pressing the **PATH** key has no effect.

PATH key (Program function mode)

Not applicable.

PATH (Alternate function mode)

Pressing the **PATH** key places the printer in the Quiet mode. The Quiet mode reduces the noise level during printing.

1.3 Printer Configuration

Introduction

The Operator Panel can be used to configure the printer. There are two unique printer configuration modes. They are:

- Power On Configuration mode
- Program Configuration mode

Power On Configuration mode

Normally accessed during the installation of the printer, the Power On Configuration mode provides complete configuration information that is based on the printer hardware and emulation firmware that are stored in the printer's Non-Volatile Memory (NVM).

Program Configuration mode

The Program Configuration mode is used to select the printer features that are in daily use. The Program Configuration mode can be selected when the printer is in the Wait state (the **ON LINE** indicator is not lit) or in the Ready state if the printer is not printing. Program information is stored in the NVM and must be accessed.

1.4 Printer Setup

Introduction

The **PROG** key is used to enter both printer Configuration modes.

To enter the Power On Configuration mode, press and hold the **PROG** key while applying power to the printer.

To enter the Program Configuration mode, press the **PROG** key while the printer is in the Wait state (the **ON LINE** indicator is not lit) or in the Ready state and not printing.

Moving within the configuration mode

There are two functional levels within each configuration mode.

- First level—Main functions
- Second level—Subfunctions for the main function

After entering the configuration mode, follow the illustrative flowchart in Figure 1-6 to move within the mode by using the left (←), right (→), up (↑), and down (↓) arrow keys on the Operator Panel.

Printer setup flowchart

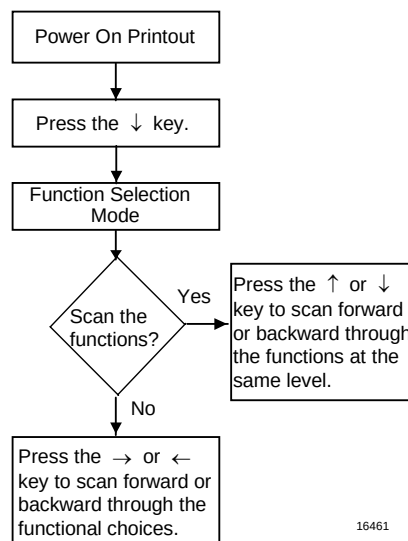


Figure 1-6 Printer Setup Flowchart

Leaving the printer configuration mode

When the Power On Configuration mode setup is complete, press the **PROG** key to exit the mode. The new setting will automatically be saved.

When the Program Configuration mode setup is complete, select one of the following display messages to determine how the new settings will be saved or ignored.

Press the → or ← key to scan forward or backward to select the desired response.

- STORE? QUIT The new settings are not stored and the old settings are still valid.
- STORE? SAVE The new settings are stored in the NVM.
- STORE? CURRENT The new settings are valid until the printer is powered down.
They are not stored in the NVM.

You can exit the configuration mode at any point in the setup process by pressing the **PROG** key.

Power On Configuration mode

Table 1-2 is a list of the Power On Configuration mode functions that can be configured.

Table 1-2 Power On Configuration Mode Functions

First Level Function	Second Level Function
Printer Paper Path Configuration	
Parallel Interface	Parallel Interface Types Select-In Signals Data Bits Buffer Sizes Emulation Types Character Sets (EPSON or IBM) National Character Sets (EPSON or IBM) Auto Carriage Return during Line Feed Auto Line Feed during Carriage Return IBM 20 CPI Compression Spacing (IBM Emulation only)
Serial Interface	Serial Interface Types Baud Rates Data Bits Parity Bits Handshaking Types Local/Remote Connections Buffer Sizes Emulation Types Character Sets (EPSON) National Character Sets (EPSON) Auto Carriage Return during Line Feed Auto Line Feed during Carriage Return (IBM Emulation only)
Functions	Front Tear Mode Rear Tear Mode Paper Overlay Sequence Selection Buzzer Quick Print Function Language Selection Ribbon color
Restore MFG	None
Sensor Tuning	None

Entering the Power On Configuration mode

Be sure that the printer power switch is in the off (0) position. While pressing and holding the **PROG** key, place the power switch in the on (1) position. Continue to hold the key until the following messages appear.

STARTING UP

■ ASPI ■ ASPI ■ ASPI ■

Release the key and the following message will be displayed.

PRINT OUT? NO

If you want to print the current configuration settings, press the → key. The configuration setting will be printed and the following message will be displayed.

PRINT OUT? YES

Examples of the default Power On Configuration mode settings are as follows.

Power On Configuration default printout

Configuration Setup Printout:

PAPER OPTIONS	PULL	NO
	ASF	NONE
PARALLEL INTERFACE	CX. PARALLEL I/F	SERIAL INTERFACE SERIAL I/F 232
	SELECT-IN	HOST BAUD 9600
	DATA BITS	8 DATA BITS 7
	DEDIC BUFFER	12K PARITY EVEN
	EMUL.	IBM XLIII ASPI (LRC) NO
	CHAR. SET	CS2 CONNECT LOCAL
	NATION	USA DEDIC BUFFER 12K
	AUTO CR	NO EMUL. ASPI
	AUTO LF	NO CHAR. SET HISE
		NATION USA
		AUTO CR NO
		AUTO LF NO
FUNCTIONS	FRONT	NORMAL
	REAR	NORMAL
	PAPER OVERLY	NO
	SEQUENCE	NONE
	BUZZER	YES
	QUICK	YES
	MENU	ENGLISH
	RIBBON	BLACK

First function of the Power On Configuration mode

If you want to display the first function (first level) of the Power On Configuration mode, press the ↓ key and the following message will be displayed.

PAPER OPTIONS

Attention

If the LOAD FRONT PUSH message is displayed because paper is inserted in the printer, the SENSOR TUNE function must be set as follows.

- Enter the Power On Configuration mode.
- Scan down the functions by pressing the ↓ key until SENSOR TUNE? NO appears on the display.
- Set the function to SENSOR TUNE? YES by pressing the → key.

Press the ↓ key to store the function and exit the Power On Configuration mode by pressing the PROG key.

Power On Configuration mode procedure menu

Table 1-3 is the setup procedure for the Power On Configuration mode menu. Use the → key to step through the second level functions in the menu and the ↓ key to store the function and move to the next first level function. The subfunction (second level) choices and their order in the menu are dependent on the final configuration that is chosen.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
PAPER OPTIONS Press → key Determines the printer's physical paper handling configuration. The manual cut sheet front path is always the standard.	PULL NO Press → key Enables the Front Push path and the Rear Push path functions. The Front Push tractor handles one paper fanfold, and the Rear Push tractor handles another paper fanfold.
	PULL YES Press → or ← key Enables the Front Pull path function (front paper entry).
	PULL PSHPL Press → or ← key Enables the Front Push and Rear Pull path function. The Front Push tractor is used together with the Rear Pull tractor to handle the same paper fanfold.
If you are at the top function of the pull path settings (PULL NO) and press the ↑ key, you will return to PAPER OPTIONS.	Press ↑ or ↓ key
The ASF functions cannot be changed if you enter directly from the CONFIG MENU YES function in the Program Configuration mode.	ASF NONE Press → or ← key ASF 1 BIN Press → or ← key ASF 2 BIN Press → or ← key ASF 3 BIN Press → or ← key
If you press the ↓ key, you move to the PARALL INTERFACE first level function.	Disables (ASF NONE) or enables (ASF 1 BIN, ASF 2 BIN, or ASF 3 BIN) the Automatic Sheet Feeder (ASF) path function.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
PARALL INTERFACE Press → key The Parallel Interface function is selected. The available second level Parallel Interface functions are as follows: Parallel Interface Type Select-in Signal Data Bits Buffer Size Emulation Character Set Nation Character Set Auto Carriage Return Auto Line Feed IBM 20 CPI Compression If you are at the beginning second level function (CX. PARALLEL I/F) for the Parallel Interface first level function and press the ↑ key, you will return to PARALL INTERFACE.	CX. PARALLEL I/F Press → key 1284 Byte I/F Press → or ← key 1284 NIBBLE I/F Press → or ← key Selects Parallel Interface type. Press ↑ or ↓ key SELECT-IN HOST Press → key SELECT-IN ON Press → or ← key Selects the SELECT-IN signal. Press ↑ or ↓ key DATA BITS 8 Press → key DATA BITS 7 Press → or ← key Selects the number of data bits. Press ↑ or ↓ key DEDIC. BUFFER 2K Press → key DEDIC. BUFFER 12K Press → or ← key DEDIC. BUFFER 32K Press → or ← key DEDIC. BUFFER 256 Press → or ← key Selects the buffer size. Press ↑ or ↓ key EMUL. ASPI Press → key EMUL. IBM XLIII Press → or ← key EMUL. IBM 2381 Press → or ← key EMUL. EPSON Press → or ← key Selects the emulation type.

First Level Function	Second Level Function
PARALL INTERFACE, continued	Press ↑ or ↓ key CHAR. SET CS2 Press → key CHAR. SET CS1 Press → or ← key CHAR. ITALIC Press → or ← key Selects the EPSON character set. Press ↑ or ↓ key CHAR. SET HISE Press → key CHAR. SET C101 Press → or ← key CHAR. SET HIS1 Press → or ← key Selects the IBM character set. Press ↑ or ↓ key NATION USA Press → key NATION FRANCE Press → or ← key NATION GERMANY Press → or ← key NATION ENGLAND Press → or ← key NATION DENMRK1 Press → or ← key NATION SWEDEN Press → or ← key NATION ITALY Press → or ← key NATION SPAIN1 Press → or ← key NATION JAPAN Press → or ← key NATION NORWAY Press → or ← key NATION DENMRK2 Press → or ← key NATION SPAIN2 Press → or ← key NATION LATIN A1 Press → or ← key NATION 8859/1 Press → or ← key NATION 8859/2 Press → or ← key NATION 8859/3 Press → or ← key NATION 8859/4 Press → or ← key NATION 8859/5 Press → or ← key NATION 8859/6 Press → or ← key NATION 8859/7 Press → or ← key NATION 8859/8 Press → or ← key NATION 8859/9 Press → or ← key Selects the EPSON nation character set.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
PARALL INTERFACE, continued	Press ↑ or ↓ key NATION USA Press → key NATION France Press → or ← key NATION Germany Press → or ← key NATION Denmark Press → or ← key NATION Norway Press → or ← key NATION Sweden Press → or ← key NATION Spain Press → or ← key NATION Finland Press → or ← key Selects the IBM nation character set. Press ↑ or ↓ key AUTO CR NO Press → key AUTO CR YES Press → or ← key Enables or disables the automatic carriage return function. Press ↑ or ↓ key AUTO LF NO Press → key AUTO LF YES Press → or ← key AUTO LF HOST Press → or ← key Selects the automatic line feed function. Press ↑ or ↓ key 20 CPI IBM NO Press → key 20 CPI IBM YES Press → or ← key Enables or disables the IBM compression function.
If you press the ↓ key, you move to the SERIAL INTERFACE first level function.	

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
SERIAL INTERFACE Press → key The Serial Interface function is selected. The second level Serial Interface functions are as follows. Interface Type Baud Rate Data Bits Parity Protocol Serial Connection Buffer Size Emulation Character Set Automatic Carriage Return Automatic Line Feed IBM 20 CPI Compressed Printing If you are at the beginning second level function (SERIAL I/F NO) for the Serial Interface first level function and press the ↑ key, you will return to the SERIAL INTERFACE first level function.	SERIAL I/F 232 Press → key SERIAL I/F 422 Press → or ← key SERIAL I/F NO Press → or ← key Selects the Serial Interface type. Press ↑ or ↓ key BAUD 9600 Press → key BAUD 19200 Press → or ← key BAUD 38400 Press → or ← key BAUD 300 Press → or ← key BAUD 600 Press → or ← key BAUD 1200 Press → or ← key BAUD 2400 Press → or ← key BAUD 4800 Press → or ← key Selects the baud rate in bits/second. Press ↑ or ↓ key DATA BITS 7 Press → key DATA BITS 8 Press → or ← key Selects the number of data bits. Press ↑ or ↓ key PARITY EVEN Press → key PARITY MARK Press → or ← key PARITY SPACE Press → or ← key PARITY NONE Press → or ← key PARITY ODD Press → or ← key Selects the parity check function. Press ↑ or ↓ key HANDSHAKE XONXOF Press → key HANDSHAKE DTR Press → or ← key Selects the handshaking function.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
SERIAL INTERFACE, continued	Press ↑ or ↓ key ASPI (LRC) NO Press → key ASPI (LRC) YES Press → or ← key Selects the handshaking function. Press ↑ or ↓ key CONNECTION LOCAL Press → key CONNECT. REMOTE Press → or ← key Selects the type of connection. Press ↑ or ↓ key DEDIC. BUFFER 12K Press → key DEDIC. BUFFER 32K Press → or ← key DEDIC. BUFFER 256 Press → or ← key DEDIC. BUFFER 2K Press → or ← key DEDIC. BUFFER 4K Press → or ← key DEDIC. BUFFER 8K Press → or ← key Selects the buffer size. Press ↑ or ↓ key EMUL. ASPI Press → key EMUL. IBM XLIII Press → or ← key EMUL. IBM 2381 Press → or ← key EMUL. EPSON Press → or ← key Selects the emulation type. Press ↑ or ↓ key CHAR. SET CS2 Press → key CHAR. SET ITALIC Press → or ← key CHAR. SET CS1 Press → or ← key Selects the EPSON character set.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
SERIAL INTERFACE, continued	Press ↑ or ↓ key CHAR. SET CS2 Press → key CHAR. SET ITALIC Press → or ← key CHAR. SET CS1 Press → or ← key Selects the EPSON character set. Press ↑ or ↓ key CHAR. SET HISE Press → key CHAR. SET C101 Press → or ← key CHAR. SET HIS1 Press → or ← key Selects the IBM character set. NATION USA Press → key NATION FRANCE Press → or ← key NATION GERMANY Press → or ← key NATION ENGLAND Press → or ← key NATION DENMRK1 Press → or ← key NATION SWEDEN Press → or ← key NATION ITALY Press → or ← key NATION SPAIN1 Press → or ← key NATION JAPAN Press → or ← key NATION NORWAY Press → or ← key NATION DENMRK2 Press → or ← key NATION SPAIN2 Press → or ← key NATION LATIN A1 Press → or ← key NATION 8859/1 Press → or ← key NATION 8859/2 Press → or ← key NATION 8859/3 Press → or ← key NATION 8859/4 Press → or ← key NATION 8859/5 Press → or ← key NATION 8859/6 Press → or ← key NATION 8859/7 Press → or ← key NATION 8859/8 Press → or ← key NATION 8859/9 Press → or ← key Selects the EPSON nation character set.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
SERIAL INTERFACE, continued	Press ↑ or ↓ key NATION USA Press → key NATION France Press → or ← key NATION Germany Press → or ← key NATION Denmark Press → or ← key NATION Norway Press → or ← key NATION Sweden Press → or ← key NATION Spain Press → or ← key NATION Finland Press → or ← key Selects the IBM nation character set. AUTO CR NO Press → key AUTO CR YES Press → or ← key Enables or disables the automatic carriage return function. Press ↑ or ↓ key AUTO LF NO Press → key AUTO LF YES Press → or ← key Enables or disables the automatic line feed function. If you press the ↓ key, you move to the FUNCTIONS first level function. If you press the ↓ key, you move to the SERIAL INTERFACE first level function.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
FUNCTIONS Press → This first level function defines the paper condition variables. The following second level functions can be defined. Front Tear Mode Rear Tear Mode Paper Overlay Sequence Buzzer Quick Language	FRONT NORMAL Press → key FRONT AUTOM. TEAR Press → or ← key FRONT LABEL Press → or ← key Selects the Front Tear mode which is either Automatic, Normal, or Label. In the Label mode there is no forward/backward movement. Press ↑ or ↓ key REAR NORMAL Press → key REAR AUTOM. TEAR Press → or ← key REAR LABEL Press → or ← key Selects the REAR TEAR mode which is either Automatic, Normal, or Label. In the Label mode there is no forward/backward movement. Press ↑ or ↓ key PAPER OVERLY NO Press → key PAPER OVERLY YES Press → or ← key Enables or disables the simultaneous MCS and tractor paper feeding. Press ↑ or ↓ key SEQUENCE NONE Press → key SEQUENCE ASF1+2 Press → or ← key SEQUENCE ASF123 Press → or ← key SEQUENCE FR+REAR Press → or ← key Selects the auto paper loading sequence which is either ASF 1+2 (first tray, then second tray), ASF 123 (first tray, then second tray, then third tray), or FR+REAR (front tractor, then rear tractor). Press ↑ or ↓ key BUZZER YES Press → key BUZZER NO Press → or ← key Enables or Disables the buzzer.

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
FUNCTIONS, continued	Press ↑ or ↓ key QUICK NO Press → key QUICK YES Press → or ← key Enables or disables quick printing. Press ↑ or ↓ key RIBBON BLACK Press → key RIBBON COLOR Press → or ← key Selects a black or color ribbon. Press ↑ or ↓ key MENU ENGLISH Press → key MENU ITALIANO Press → or ← key MENU FRANCAIS Press → or ← key MENU ESPANA Press → or ← key MENU DEUTSCH Press → or ← key Selects the Operator panel language. If you press the ↓ key, you move to the BACK TO MFG:NO first level function.
BACK TO MFG: NO Press → key BACK TO MFG: YES Press ← key Select the BACK TO MFG: YES function, then press ↑ or ↓ key, to restore the factory default function values for Power On Configuration setup and Program Configuration setup. Select the BACK TO MFG: NO function to store the newly selected function values for Power On Configuration setup and Program setup when exiting the Power ON Configuration setup mode.	None

Table 1-3 Power on Configuration Mode Procedure Menu

First Level Function	Second Level Function
SENSOR TUNE? NO Press → key SENSOR TUNE? YES Press ← key Select the SENSOR TUNE? YES function, then press ↑ or ↓ key, to enable paper path tuning. The function must be enabled when no paper is inserted in any of the paper paths. This function is normally performed by a technician after the main electronics board is replaced or if the loaded paper was not recognized.	None

At this point, the Power On Configuration mode setup procedure is complete. To exit the menu, press the **PROG** key and the new setup will be automatically saved.

Program Configuration mode

Table 1-4 is a list of the Program Configuration mode functions that can be setup.

Table 1-4 Program Configuration Mode Functions

First Level Function	Second Level Function
Form Path	None
Paper Path	None
Tuning Selection *	None
Gap Tuning	Horizontal Tuning Vertical Tuning
Form Settings *	Line Spacing Lines Per Inch Lock Page Length Top of Form Skipover Font Pitch Characters Per Second Lock Left and Right Margins Slashed Zero Return to MFG
Another Fault Detection	None
Power On Configuration Setup	None
Hexadecimal Printing	None
Store Mode	None
* The marked functions configure only the paper path that is selected in the Paper Path function.	

Entering the Program Configuration mode

While the printer is in the Wait state or in the Ready state and not printing, press the **PROG** key. The following message will appear on the display.

PRINT OUT? NO

If you wish to print the current Program Configuration mode setup, press the → key. The following message will appear on the display.

PRINT OUT? YES

An example of a printout for a default configuration setup follows. When printing is complete, the following message will appear on the display.

PRINT OUT? NO

Default Program Configuration printout

Program Setup Printout:

1 FORM PER PATH

PATH MANUAL FORM	PATH PUSH PULL
AUTOGAP 0	AUTOGAP 0
TUNING: HORIZ 0	TUNING: HORIZ 0
TUNING: VERT 0	TUNING: VERT 0
LINE SP. 6 LPI	LINE SP. 6 LPI
LINE SP. LOCK NO	LINE SP. LOCK NO
LENGTH A4	LENGTH 66 LINES
TOP OF FORM 0	TOP OF FORM 0
SKIPOVER 0	SKIPOVER 0
FONT DRAFT	FONT DRAFT
PITCH 10 CPI	PITCH 10 CPI
PITCH LOCK NO	PITCH LOCK NO
LEFT MARGIN 0	LEFT MARGIN 0
RIGHT MARGIN 80	RIGHT MARGIN 136
SPLASH ZERO NO	SPLASH ZERO NO

Medium Program Configuration mode setup

An example of a printout for a medium configuration setup follows.

Medium Program Configuration printout

-Program Setup Printout:

1 FORM PER PATH

PATH MANUAL FORM	PATH FRONT PUSH	PATH REAR PUSH
AUTOGAP 0	AUTOGAP 0	AUTOGAP 0
TUNING: HORIZ 0	TUNING: HORIZ 0	TUNING: HORIZ 0
TUNING: VERT 0	TUNING: VERT 0	TUNING: VERT 0
LINE SP. 6 LPI	LINE SP. 6 LPI	LINE SP. 6 LPI
LINE SP. LOCK NO	LINE SP. LOCK NO	LINE SP. LOCK NO
LENGTH A4	LENGTH 66 LINES	LENGTH 66 LINES
TOP OF FORM 0	TOP OF FORM 0	TOP OF FORM 0
SKIPOVER 0	SKIPOVER 0	SKIPOVER 0
FONT DRAFT	FONT DRAFT	FONT DRAFT
PITCH 10 CPI	PITCH 10 CPI	PITCH 10 CPI
PITCH LOCK NO	PITCH LOCK NO	PITCH LOCK NO
LEFT MARGIN 0	LEFT MARGIN 0	LEFT MARGIN 0
RIGHT MARGIN 80	RIGHT MARGIN 136	RIGHT MARGIN 136
SPLASH ZERO NO	SPLASH ZERO NO	SPLASH ZERO NO

Entering the Program Configuration setup procedure menu

To enter the Program Configuration mode procedure menu, press the ↓ key and the following message will be displayed.

1 FORM PER PATH

You have entered the Program Configuration mode procedure menu that is listed in Table 1-5.

Program Configuration mode procedure menu

Table 1-5 is the setup procedure for the Program Configuration mode menu. Use the → key to step through the menu and the ↓ key to store the function.

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
1 FORM PER PATH When you load the paper, the paper path is defined by the default associated parameters that were selected by the menu or by firmware. Press → key ONE FORM ONLY When you load the paper, the paper path is defined by the current associated parameters and not the default parameters selected by the menu or by firmware. Press the → or ← key to return to the 1 FORM PER PATH function. If you press the ↓ key, you move to the PATH FRONT PUSH first level function .	None
PATH FRONT PUSH Press → or ← key Paper is fed by the front push tractor. Press → or ← key	None
PATH MANUAL FORM Manual cut sheet feeding is the standard. Press → or ← key	None

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
PATH FRONT PULL Paper is fed by the rear pull tractor. Selecting this paper path automatically excludes the PATH FRONT PULL and PATH PUSH PULL paper paths. Press → or ← key	None
PATH FRONT PUSH Paper is fed by the front pull tractor. Selecting this paper path automatically excludes the PATH REAR PULL and PATH PUSH PULL paper paths. Press → or ← key	None
PATH PUSH PULL Paper is fed by the rear push/pull tractor. Selecting this paper path automatically excludes the PATH REAR PULL and PATH FRONT PULL paper paths. If you press the ↑ key, you move back to the 1 FORM PER PATH first level function. If you press the ↓ key, you move to the TUNING first level function.	None

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
TUNING Press → key	AUTO GAP 0 Press → key
The following tuning functions are available.	AUTO GAP +1 Press → or ← key
Gap Tuning	AUTO GAP +2 Press → or ← key
Horizontal Tuning	AUTO GAP +3 Press → or ← key
Vertical Tuning	MANUAL GAP Press → or ← key
If you are at the top second level tuning function (MANUAL GAP 0) press the ↑ key to move back to the TUNING first level function.	FIXED GAP 0 Press → or ← key
	FIXED GAP 1 Press → or ← key
	FIXED GAP 2 Press → or ← key
	FIXED GAP 3 Press → or ← key
	FIXED GAP 4 Press → or ← key
	FIXED GAP 5 Press → or ← key
	FIXED GAP 6 Press → or ← key
	FIXED GAP 7 Press → or ← key
	AUTO GAP -5 Press → or ← key
	AUTO GAP -4 Press → or ← key
	AUTO GAP -3 Press → or ← key
	AUTO GAP -2 Press → or ← key
	AUTO GAP -1 Press → or ← key
	Selects the paper gap distance for a fixed (from 0 to 7) or automatic (from -5 to +3) amount or manual.
	Press ↑ or ↓ key
	TUNING: HORIZ 0...60 Press → or ← key
	Determines the distance of the first printable character from the left paper margin. The range of the values are from -0.5 to +0.5 inch in 1/120-inch increments.
	Press ↑ or ↓ key
	TUNING: VERT -30...30 Press → or ← key
	Determines the distance of the first printable character from the top paper margin. The range of the values are from -1/16 to +1/6 inch in 48 1/144-inch increments.
	If you press the ↓ key, you move back to the TUNING first level function.
If you press the ↓ key, you move to the FORM SETTING first level function.	

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
FORM SETTING Press → key The following form setting functions are available. Line Spacing Lines Per Inch Lock Page Length Top of Form Skipover Font Pitch Characters Per Inch Lock Left Margin Right Margin Slashed Zero Store Mode If you are at the top second level Tuning function (LINE SP. 6 LPI) press the ↑ key to move back to the FORM SETTING first level function.	LINE SP. 6 LPI Press → key LINE SP. 8 LPI Press → or ← key LINE SP. 12 LPI Press → or ← key LINE SP. 3L/30MM Press → or ← key LINE SP. 4L/30MM Press → or ← key LINE SP. 6L/30MM Press → or ← key LINE SP. 8L/30MM Press → or ← key LINE SP. 12L/30MM Press → or ← key Selects the vertical spacing in lines per inch (6, 8, 12) or in lines per 30 mm (3, 4, 6, 8,12). Press ↑ or ↓ key LINE SP. LOCK NO Press → key LINE SP. LOCK YES Press → or ← key The vertical spacing setting can be overridden by the firmware (No) or by the Operator Panel (Yes). Press ↑ or ↓ key LENGTH A4 Press → key LENGTH A3 Press → or ← key LENGTH A2 Press → or ← key LENGTH LEGAL Press → or ← key LENGTH LETTER Press → or ← key LENGTH xxx LINES Press → or ← key LENGTH A5 Press → or ← key Selects the fanfold paper length in the number of lines, and for the cut sheets as either a standard format (letter, legal, A2, A3, A4, A5) or the number of lines. Press ↑ or ↓ key TOP OF FORM xxx Press → or ← key Defines the top of form in number of lines (1/16 inch).

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
FORM SETTING, continued	Press ↑ or ↓ key SKIP OVER xxx Press → or ← key Defines the skipover function in the number of lines. Press ↑ or ↓ key FONT Draft Press → key FONT Courier Press → or ← key FONT OCR-B Press → or ← key FONT Gothic Press → or ← key FONT OCR-A Press → or ← key Press ↑ or ↓ key PITCH 10 CPI Press → key PITCH 12 CPI Press → or ← key PITCH 15 CPI Press → or ← key PITCH 16.7 CPI Press → or ← key PITCH 5 CPI Press → or ← key PITCH 6 CPI Press → or ← key PITCH 7.5 CPI Press → or ← key PITCH 8.4 CPI Press → or ← key Determines the horizontal spacing in characters per inch (CPI). Press ↑ or ↓ key PITCH LOCK NO Press → key PITCH LOCK YES Press → or ← key Determines whether the horizontal spacing can be overridden by firmware (No) or by the Operator Panel (Yes).

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
FORM SETTING, continued	Press ↑ or ↓ key LEFT MARGIN xxx Press → or ← key Determines the left margin in 1/10-inch blank columns starting from the edge of the paper. Press ↑ or ↓ key RIGHT MARGIN xxx Press → or ← key Determines the right margin in 1/10-inch blank columns starting from the right edge of the paper. Press ↑ or ↓ key SLASH ZERO NO Press → key SLASH ZERO YES Press → or ← key Enables (Yes) or disables (No) the slashed zero function. Press ↑ or ↓ key RETURN → MFG NO Press → key RETURN → MFG YES Press → or ← key Return to the factory default settings for the selected paper path (Yes) or save the new settings in the NVM (No). If you press the ↓ key, you move back to the FORM SETTING first level function.
You have completed the configuration of the selected paper path. To configure another paper path, press the ↓ key to move to the ANOTHER PATH NO first level function.	

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
ANOTHER PATH NO This selection does not allow you to configure another default paper path. Press → or ← key ANOTHER PATH YES This selection allows you to configure another default paper path. If you press the ↓ key, you move forward to the CONFIG MENU NO first level function.	None
CONFIG MENU NO This selection does not allow you to return to the Power On Configuration mode. Press → key CONFIG MENU YES This selection allows you to return to the Power On Configuration mode. If you press the ↑ key, you move to the PRINT OUT? first level function in the Power On Configuration mode. If you press the ↓ key, you move forward to the HEX DUMP NO first level function.	None

Table 1-5 Program Configuration Mode Procedure Menu

First Level Function	Second Level Function
HEX DUMP NO This selection does not select the hexadecimal dump printing. Press → key HEX DUMP YES This selection selects the hexadecimal dump printing. If you press the ↑ key, you return to the CONFIG MENU NO first level function. If you press the ↓ key, you move forward to the STORE QUIT first level function.	None
STORE? QUIT This selection ignores the new Program Configuration mode settings. The previous settings are still valid. If you press the → key, you move forward to the STORE SAVE first level function. STORE? SAVE This selection stores the new Program Configuration mode settings in the NVM. The new settings are now valid. If you press the → key, you move forward to the STORE CURRENT first level function. STORE? CURRENT The new Program Configuration mode settings are valid only until power is removed from the printer.	None

At this point, the Program Configuration mode setup procedure is complete. To exit the menu, press the **PROG** key.

1.5 Paper Handling

Paper paths

Figure 1-7 illustrates all the paper paths that can be configured when the optional second tractor is installed in the printer.

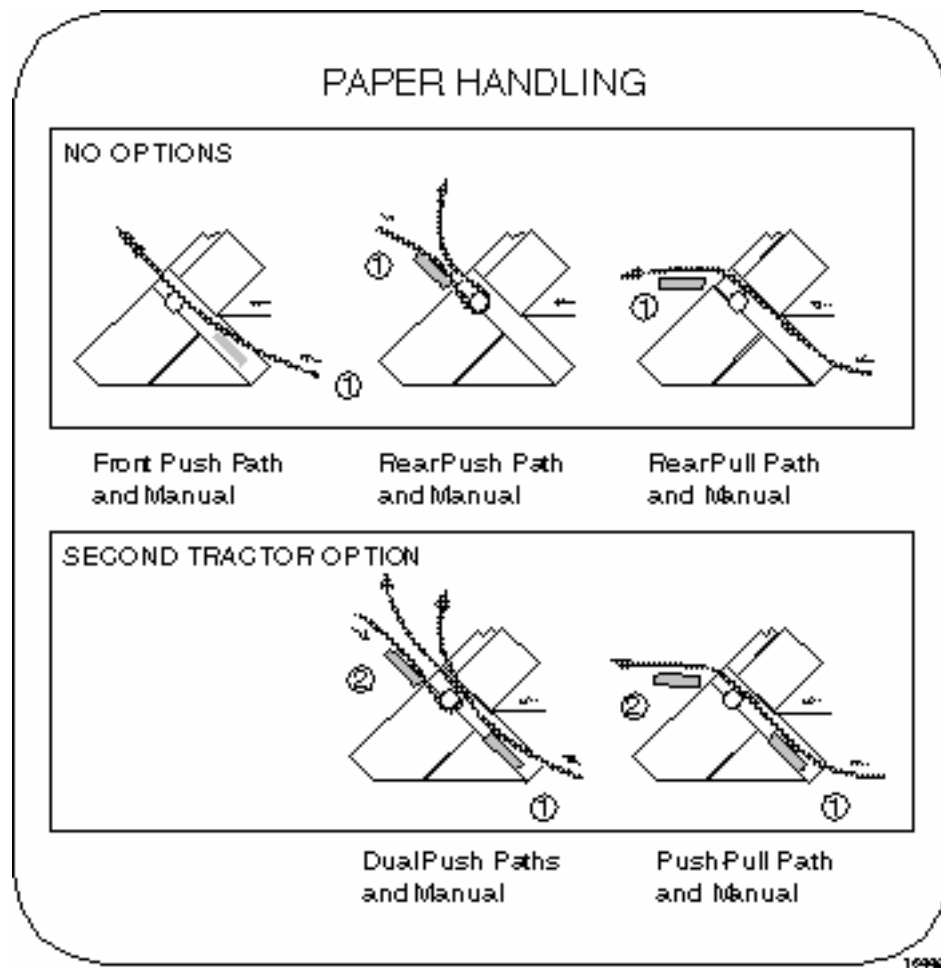


Figure 1-7 Paper Handling Selections

Paper specifications

It is important to use the correct paper to obtain the best performance. See Tables 1-6 and 1-7.

Table 1-6 Fanfold Paper Specifications

Path	Width (Inches)	Height (Inches)	Maximum Thickness (Millimeters)	Copies
Front Push Path	3 to 17.5	3 to 24	0.64	1 + 7
Front Pull Path (Front Feed)	3 to 17.5	3 to 24	0.64	1 + 7
Front Pull Path (Bottom Feed)	3 to 15.0	3 to 24	0.64	1 + 7
Rear Push Path	3 to 17.5	3 to 24	0.35	1 + 3

Table 1-7 Fanfold Paper Weight

Paper Type	Paper Weight	
Original	55-80 g/m ² (Rear Push)	
Original	55-150 g/m ² (Front Push and Rear Pull)	
Multipart Form	Sheets	44-75 g/m ²
Multipart Form	Carbon	35 g/m ²

The paper parameters that are set by the Operator panel or by software are usually equal to the physical parameters.

1.6 Fanfold Paper Path Configurations

Introduction

There are five fanfold paper path configurations. The Dual Front-Rear Push paths and Front Push-Rear Pull path configurations require that a second option tractor be installed on the printer.

Front Push paper path

The Front Push paper path configuration provides continuous feed of fanfold paper from the front to the rear of the printer. A single tractor must be installed at the front of the printer. Paper is pushed through the printer.

Rear Push paper path

The Rear Push paper path configuration provides continuous feed of fanfold paper from the rear of the printer to the rear of the printer. A single tractor must be installed at the rear of the printer. Paper is pushed through the printer.

Rear Pull paper path

The Rear Pull paper path configuration provides continuous feed of fanfold paper from the front to the rear of the printer. A single tractor must be installed at the rear of the printer. Paper is pulled through the printer.

Front Push-Rear Pull paper path

The Front Push-Rear Pull paper path configuration provides continuous feed of fanfold paper from the front to the rear of the printer. Tractors must be installed at both the front and rear of the printer. Paper is pushed and pulled through the printer.

Dual Front-Rear Push paper paths

The Dual Front-Rear Push paper path configuration provides alternate feed of fanfold paper at both the front and rear of the printer. Both paper paths exit the rear of the printer. Tractors must be installed at both the front and rear of the printer. Paper is pushed through the printer.

1.7 Front Push Paper Path Configuration

Paper loading procedure

Use the procedure in Table 1-8 to load the fanfold paper for a Front Push paper path configuration. The appropriate settings must be selected in the Program Configuration.

Table 1-8 Front Push Paper Path Loading Procedure

Step	Action
1	If the single sheet support cover is open, close the cover by pulling it upward before loading the fanfold paper.
2	Press the PATH key on the Operator panel to select the Front Push paper path. The Front Push Fanfold Path LED will light, and the LOAD PAPER: FRONT message will appear on the display.
3	Lift the front tractor cover and fasten it to the single sheet support cover with its magnetic lock.
4	Unlock the tractors by placing the levels in the up position. Position the left tractor on the first print column.
5	Align the paper guides on the tractor bar. Open the tractor covers.
6	Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
7	Lock the tractors by placing the levers in the down position. Close the tractor covers.
8	Close the front tractor cover by pushing it downward.
9	Press the LOAD/FF key on the Operator panel to register the paper.

1.8 Rear Push Paper Path Configuration

Paper loading procedure

Use the procedure in Table 1-9 to load the fanfold paper for a Rear Push paper path configuration. The appropriate settings must be selected in the Program Configuration.

When you select this paper path, the Rear Pull and Front Push-Rear Pull paths are excluded.

Table 1-9 Rear Push Paper Path Loading Procedure

Step	Action
1	Press the PATH key on the Operator Panel to select the Rear Push paper path. The Rear Push Fanfold Path LED will light, and the LOAD PAPER: REAR message will appear on the display.
2	Place the Push/Pull slider in the PUSH position (toward the rear of the printer) and the rear tractor in the up position. Figure 1-8 illustrates the proper positioning.
3	Lift the paper separator that is located on the tractor unit.
4	Unlock the tractors by placing the levers in the down position. Position the left tractor on the first print column.
5	Align the paper guides on the tractor bar. Open the tractor covers.
6	Pass the paper under the paper separator. Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
7	Lock the tractors by placing the levers in the up position. Close the tractor covers.
8	Press the LOAD/FF key on the Operator Panel to register the paper.

Push/Pull slider in push position

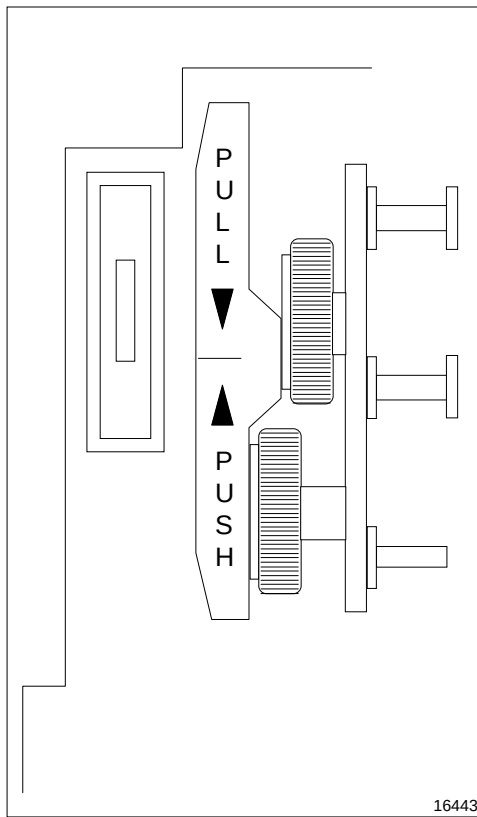


Figure 1-8 Push/Pull Slider in Push Position

1.9 Rear Pull Paper Path Configuration

Paper loading procedure

Use the procedure in Table 1-10 to load the fanfold paper for a Rear Pull paper path configuration. The appropriate settings must be selected in the Program Configuration.

When you select this paper path, the Rear Push and Front Push-Rear Pull paths are excluded.

Table 1-10 Rear Pull Paper Path Loading Procedure

Step	Action
1	Press the PATH key on the Operator Panel to select the Rear Pull paper path. The LOAD PAPER: PULL message will appear on the display and the print head will move to the rightmost position on the carriage.
2	Place the Push/Pull slider in the PULL position (towards the front of the printer). Figure 1-9 illustrates the proper positioning of the slider.
3	Unlock the levers that are located under the rear tractor unit by placing them in the down position. Carefully press the tractor unit downward until it clicks into the pull position.
4	Open the top cover and carefully lift the paper bail. Be sure the print head is positioned to the extreme right on the carriage. Close the top cover.
5	Lift the front tractor cover and fasten it to the single sheet support cover with its magnetic lock. Insert the paper through the front paper entry slot.
6	Unlock the rear tractors by placing the levers in the down position. Position the left tractor on the paper's first print column.
7	Align the paper guides on the tractor bar. Open the tractor covers.
8	Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
9	Lock the tractors by placing the levers in the up position. Close the tractor covers.
10	Press the ON LINE key on the Operator Panel to confirm that the paper is properly loaded.

Push/Pull slider in pull position

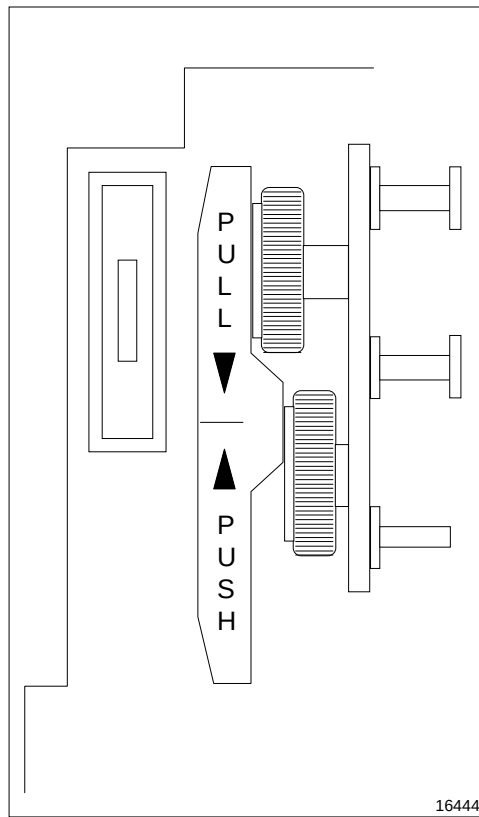


Figure 1-9 Push/Pull Slider in Pull Position

1.10 Rear Pull Paper Path Configuration – Bottom Load

Paper loading procedure

Use the procedure in Table 1-11 to load the fanfold paper for a Rear Pull paper path configuration. The appropriate settings must be selected in the Program Configuration.

When you select this paper path, the Rear Push and Front Push-Rear Pull paths are excluded. The optional printer pedestal or equivalent furniture must be used for this paper path.

Table 1-11 Rear Pull Paper Path Loading Procedure – Bottom Load

Step	Action
1	Press the PATH key on the Operator Panel to select the Rear Pull paper path. The LOAD PAPER: PULL message will appear on the display and the print head will move to the rightmost position on the carriage.
2	Place the Push/Pull slider in the PULL position (towards the front of the printer). Figure 1-9 illustrates the proper positioning of the Push/Pull slider.
3	Unlock the levers that are located under the rear tractor unit by placing them in the down position. Carefully press the tractor unit downward until it clicks into the pull position.
4	Open the top cover and carefully lift the paper bail. Be sure the print head is positioned to the extreme right on the carriage. Close the top cover.
5	Unlock the rear tractors by placing the levers in the down position. Position the left tractor on the paper's first print column. Align the paper guides on the tractor bar.
6	Insert the paper through the bottom entry slot.
7	Open the tractor covers.
8	Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
9	Lock the tractors by placing the levers in the up position. Close the tractor covers.
10	Press the ON LINE key on the Operator Panel to confirm that the paper is properly loaded.

1.11 Front Push-Rear Pull Paper Path Configuration

Paper loading procedure

Both front and rear tractor units are used in this paper path configuration. The fanfold paper is loaded onto the Front Push tractor unit and exits on the Rear Pull tractor unit to provide a straight and fast paper output.

Use the procedure in Table 1-12 to load the fanfold paper for a Push-Pull paper path configuration. The appropriate settings must be selected in the Program Configuration.

When you select this paper path, the Rear Push and the Rear Pull paths are excluded.

Table 1-12 Front Push-Rear Pull Paper Path Loading Procedure

Step	Action
1	If the single sheet support cover is open, close the cover by pulling it upward before loading the fanfold paper.
2	Press the PATH key on the Operator Panel to select the Front Push-Rear Pull paper path. The Front Push Fanfold Path LED will light, and the LOAD PAPER: PSHPL message will appear on the display.
3	Place the Push/Pull slider in the PULL position (towards the front of the printer). Figure 1-9 illustrates the proper positioning of the slider.
4	Lift the front tractor cover and fasten it to the single sheet support cover with its magnetic lock.
5	Unlock the tractors by placing the levers in the up position. Position the left tractor on the first print column.
6	Align the paper guides on the tractor bar. Open the tractor covers.
7	Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
8	Lock the tractors by placing the levers in the down position. Close the tractor covers.
9	Press the LOAD/FF key on the Operator Panel to register the paper.
10	Close the front tractor cover by pushing it downward.
11	Unlock the rear tractors by placing the levers in the down position. Position the left tractor on the papers first print column.
12	Align the paper guides on the tractor bar. Open the tractor covers.

Table 1-12 Front Push-Rear Pull Paper Path Loading Procedure

Step	Action
13	Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
14	Lock the tractors by placing the levers in the up position. Close the tractor covers.
15	Press the ON LINE key on the Operator Panel to confirm that the paper is properly loaded.

1.12 Dual Front-Rear Push Paper Path Configuration

Paper loading procedure

Both front and rear tractor units are used in this dual paper path configuration. Fanfold paper is loaded onto the Front Push tractor unit and exits at the rear of the printer. A second source of fanfold paper is loaded onto the Rear Push tractor unit from the rear of the printer and also exits to the rear of the printer. Paper movement can be alternated between the two paper paths.

Use the procedure in Table 1-13 to load the fanfold paper for a Dual Front-Rear Pull paper path configuration. The appropriate settings must be selected in the Program Configuration.

Table 1-13 Dual Front-Rear Push Paper Path Loading Procedure

Step	Action
1	Place the Push/Pull slider in the PUSH position (toward the rear of the printer) and the rear tractor in the up position. Figure 1-8 illustrates the proper positioning.
2	If the single sheet support cover is open, close the cover by pulling it upward before loading the fanfold paper.
3	Press the PATH key on the Operator Panel to select the Front Push paper path. The Front-Push Fanfold Path LED will light, and the LOAD PAPER: FRONT message will appear on the display.
4	Lift the front tractor cover and fasten it to the single sheet support cover with its magnetic lock.
5	Unlock the tractors by placing the levers in the up position. Position the left tractor on the first print column.
6	Align the paper guides on the tractor bar. Open the tractor covers.
7	Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
8	Lock the tractors by placing the levers in the down position. Close the tractor covers.
9	Close the front tractor cover by pushing it downward.
10	Press the LOAD/FF key on the Operator Panel to register the paper.

Table 1-13 Dual Front-Rear Push Paper Path Loading Procedure

Step	Action
11	Press the PATH key on the Operator Panel to select the Rear Push paper path. The Rear-Push Fanfold Path LED will light, and the LOAD PAPER: REAR message will appear on the display.
12	Place the Push/Pull slider in the PUSH position (toward the rear of the printer) and the rear tractor in the up position. Figure 1-8 illustrates the proper positioning.
13	Lift the paper separator that is located on the tractor unit.
14	Unlock the tractors by placing the levers in the down position. Position the left tractor on the first print column.
15	Align the paper guides on the tractor bar. Open the tractor covers.
16	Pass the paper under the paper separator. Load the paper by aligning the paper holes with the tractor pins of the left and right tractor set.
17	Lock the tractors by placing the levers in the up position. Close the tractor covers.
18	Press the LOAD/FF key on the Operator Panel to register the paper.

2. Signum 2043 Printer Service

2.1 Overview

Introduction

The Signum 2043 matrix printer is a replacement for the ASPI-32 and ASPI-41 printers. It is a 136-column printer, capable of 400 characters per second with an APSI protocol. Seven printable colors are available when a color ribbon is installed.

Two models of the printer are available, a 120 Vac model and a 240 Vac model. Both models are CE Compliant.

Illustrations

Figures 2-1 and 2-2 are front and rear views of the Signum 2043 printer. Figure 2-3 is a pictorial of the printer's components.

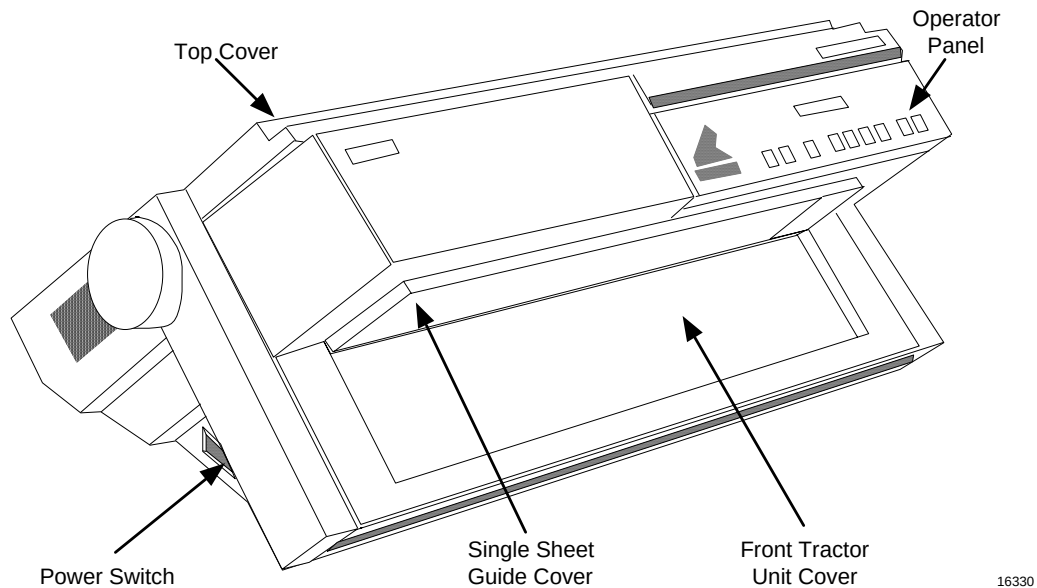


Figure 2-1 Front View of Signum 2043 Printer

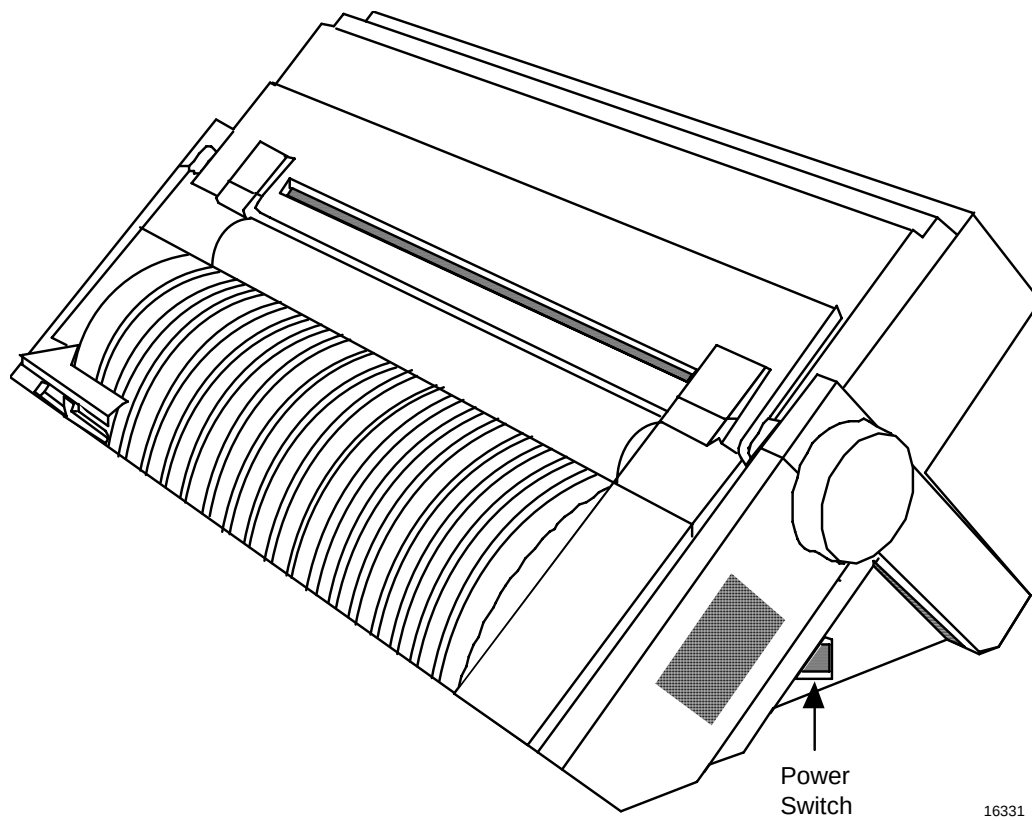


Figure 2-2 Rear View of Signum 2043 Printer

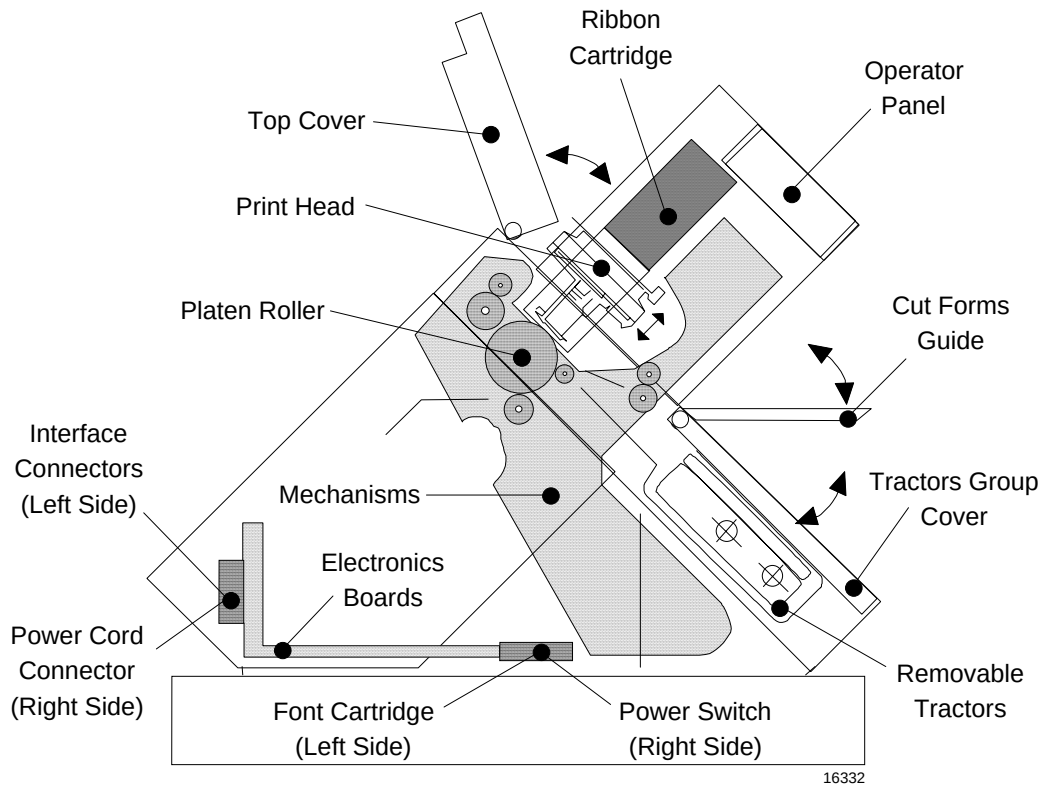


Figure 2-3 Signum 2043 Printer Pictorial

2.2 Printer Cleaning

Attention

Due to the high temperature of some mechanical parts during prolonged operation (print head, carriage motor, paper motor), be sure power has been removed from the printer for at least 5 minutes before beginning any cleaning.

Cleaning procedure

There is no scheduled preventive maintenance other than periodic cleaning. Periodic cleaning of the printer will help provide optimal printer performance. Use the printer cleaning procedure in Table 2-1.

Table 2-1 Printer Cleaning Procedure

Step	Action
1	Place the printer's power switch that is located at the left side of printer in the OFF position and unplug the power cord.
2	Open the top cover. If paper is loaded into the printer, remove the paper.
3	With a vacuum cleaner, remove any dirt or dust inside the printer. If necessary, use a brush to loosen any dirt accumulation, but be careful not to damage or misalign delicate components, such as the print head.
4	Close the cover and clean the exterior of the printer by using a soft-bristled brush or vacuum cleaner. Use a neutral detergent or water solution on a soft cloth to remove dirt and grease from the printer cabinet. Do not use an abrasive cloth, alcohol, paint thinner, or similar agents because they may discolor or scratch the cabinet surface.
5	Reconnect the power cord to the printer and place the printer's power switch in the ON position.

2.3 Printer Adjustments

No adjustments

There are no printer adjustments at the ORU (Optimum Replaceable Unit) level.

2.4 Troubleshooting/Testing

Introduction

To help you isolate and correct minor printer problems, the following problem solving checks and procedures are recommended.

- Initializing procedure with Operator Panel messages
- Power malfunction checks
- Self-test procedure
- Test and diagnostic (T&D) procedure with error messages

Troubleshooting approach

When your printer has a problem, use the general troubleshooting/testing procedure in Table 2-2.

Table 2-2 General Troubleshooting Procedure

Step	Action
1	If appropriate, perform the power malfunction checks.
2	If appropriate, observe the initialization procedure and Operator Panel messages.
3	Perform the printer self-test procedure. The instructions are in subsection 2.11.
4	Perform the diagnostic procedure. The instructions are in subsection 2.14.
5	If unable to diagnose and correct a problem, call the Honeywell Technical Assistance Center (TAC) for assistance. • In the U.S.A., use the toll-free number 1*800-822-7673 (available in the contiguous states except Arizona). • In Arizona, use 602-313-5558.

2.5 Power Malfunctions

WARNING

The printer power supply is a switching type with some parts exposed to high voltage. The power supply cover must be in place before performing any operation while the printer's rear cover is removed.

Problem Solving

Use the suggestions in Table 2-3 to help identify and correct a power malfunction.

Table 2-3 Power Malfunction Guide

Problem	Possible Cause	Solution
POWER indicator extinguished	Power switch not in the on (1) position.	Place the printer's power switch in the on (1) position.
	Power cable not connected.	Verify that the cable is properly connected to the printer and the ac power outlet.
	Power absent at ac power outlet.	Verify that power is present at the ac power outlet.
	Primary power fuse blown.	Replace the primary power fuse. 120 Vac – 5 A (1 fuse) 240 Vac – 3.15 A (2 fuses)
No display information	The Operator Panel assembly is faulty or has a bad connection.	Check the connection to the main circuit board assembly.

2.6 Printer Initialization

Introduction

When power is applied to the printer by placing the Power switch in the on (1) position, it enters the initialization mode (bootstrap) before entering the normal operational mode. While in the initialization mode, the printer firmware initializes the hardware resources for the correction functionality.

The printer display will momentarily display: **STARTING UP** and then perform a full-matrix display check from the leftmost to the rightmost character position.

The printer display will momentarily display: ■ ■ ■ ■ ■ ■ ■ ■ ■ ■

If initialization is successful, the printer enters the normal operational mode, moves the carriage assembly to the leftmost position to check the home position sensor, and then enters either the READY (paper installed) or WAIT (paper not installed) state.

If the printer does not initialize correctly, an error message is displayed on the printer display. See subsections 2.7 through 2.9 for lists of error messages.

2.7 Printer Display Errors

Introduction

Some errors can be displayed by the printer. The errors are divided into three main groups.

- Status errors
- Recoverable errors
- Nonrecoverable errors

2.8 Status Errors

Introduction

When a Status error occurs, the printer enters the WAIT state and displays a primary message that indicates the error. If appropriate, a secondary error message is displayed that provides more specific information about the primary error.

Pressing the ON LINE key resets the error condition.

Error messages

Table 2-4 describes the printer status error messages.

Table 2-4 Printer Status Errors

First Message	Second Message	Cause/Solution
PAPER JAM	CHECK ALL PATHS	Cause: A paper jam error condition exists in the paper path. Solution: Check all the paper paths and remove the jammed paper. Press the ON LINE key to reset the error condition.
JAM ON PATH2	CHECK PAPER	Cause: A paper jam error condition exists in paper path 2. Solution: Check the front push tractor unit and remove the jammed paper. Press the ON LINE key to reset the error condition.
JAM ON PATH3	CHECK PAPER	Cause: A paper jam error condition exists in paper path 3. Solution: Check the front push tractor unit and remove the jammed paper. Press the ON LINE key to reset the error condition.
AGA NOT OPER.	ADJUST THE GAP	Cause: The automatic gap adjustment (A.G.A.) is not enabled. Solution: Manually adjust the gap. Press the ON LINE key to reset the error condition.
BUFFER OVERFLOW	CHARACTER LOST	Cause: An overflow condition exists in the buffer. Solution: Place the printer in the OFF position, and the buffer will be cleared. Press the ON LINE key to reset the error condition.

Table 2-4 Printer Status Errors

First Message	Second Message	Cause/Solution
DATA SET OFF	N/A	Cause: The DSR signal is off, and the printer is not ready for data transfer. Solution: Press the ON LINE key to reset the error condition.
RIBBON BLOCKED	CHECK RIBBON	Cause: The cartridge ribbon is blocked. Solution: Check to be sure that the ribbon cartridge is inserted correctly. Turn the tension knob to verify that the ribbon is not jammed. Press the ON LINE key to reset the error condition.
PATH2 TRIM ERROR	N/A	Cause: A trimmer sensor error exists in paper path 2. Solution: Press the ON LINE key to reset the error condition.
PATH3 TRIM ERROR	N/A	Cause: A trimmer sensor error exists in paper path 3. Solution: Press the ON LINE key to reset the error condition.
PRINT INTEGRITY	N/A	Cause: Abnormal print out because of a possible print carriage blocking. Solution: Press the ON LINE key to reset the error condition.

2.9 Recoverable Errors

Introduction

When a recoverable error occurs, the printer enters the WAIT state and displays a primary error message. If appropriate, a secondary error message is displayed that provides more specific information about the primary error. An intermittent buzzer may sound.

All error displays are reset by either pressing the ON LINE key or by a specific action.

Error messages

Table 2-5 describes the printer status error messages.

Table 2-5 Printer Recoverable Errors

First Message	Second Message	Cause/Solution
# COVER OPEN	N/A (Continuous buzzer)	Cause: The top cover was open during the Initialization mode. Solution: Close the cover. Check to see if a magnet is installed on the cover. Cause: The Operator Panel may be faulty. Solution: Replace the Operator Panel.
CLOSE COVER	N/A	Cause: The top cover was opened during operation. Solution: Close the top cover.
PAPER JAM	CHECK ALL PATHS	Cause: A paper jam error condition existed in a paper path during a parking or paper ejection operation. Solution: Check to see that the paper is correctly installed and remove the paper if it is jammed.
RIBBON BLOCKED	CHECK RIBBON	Cause: The cartridge ribbon is blocked and the relative sensors detect the fault. The problem occurs when some characters are printed with poor ink density. Solution: Check that the ribbon is properly inserted. To be sure that the ribbon is not jammed, remove the ribbon and turn the tension knob.

Table 2-5 Printer Recoverable Errors

First Message	Second Message	Cause/Solution
PRINT INTEGRITY	N/A	Cause: The carriage assembly is blocked and the relative home position sensor detected the fault. The problem occurs when the printout is shifted to the right or left side. Solution: Check to see that the carriage assembly is free.
BUFFER OVERFLOW	CHARACTER LOST	Cause: An overflow condition occurred in the input buffer. The EPDG I/O or EPDGC I/O board is transmitting data to the printer even though it was signaled to stop. Solution: Check the connection between the printer and the EPDGC I/O board. Press the ON LINE key to reset the error. If the buffer must be cleared, cycle the printer power off, then back on.
DATA SET OFF	N/A	Cause: The DSR signal to the printer is not active, and the printer is not ready for data transfer. Solution: Check the connection between the printer and the EPDGC I/O board.

2.10 Nonrecoverable Errors

Introduction

When a nonrecoverable error occurs, the printer is halted and enters an Error state. A nonrecoverable error cannot be reset by a specific action, and therefore the printer power must be cycled off and back on.

Typically, a nonrecoverable error is detected while the printer in the Bootstrap or normal operational state.

If the nonrecoverable error repeats after cycling the printer power off and back on, run the T&D test that is described in subsection 2.14 to identify the defective ORU. If after correction or replacement of the identified ORU, the nonrecoverable error reoccurs, contact your nearest Honeywell Technical Assistance Center.

Bootstrap error messages

In the Bootstrap state, the following error messages could be displayed. If one of the messages does appear, the Bootstrap will not complete, and consequently the printer hardware is not properly initialized.

- STARTING UP (fixed)
- SOFTWARE ERROR
- ENGINE DRIVER
- PM SET-UP ERR
- # VOID EXEC LIST
- # MFILE CHECKSUM
- UNK MFILE TYPE
- TOO MANY FILES
- # CARD 0 REMOVED

Generally, all the above error messages are related to either an EPROM, microcontroller, or data bus failure.

Normal operation error messages

In the normal operational mode, the following messages may be displayed. If one of the messages does appear, the Bootstrap was completed, but a handshaking problem occurred between two of the firmware modules.

- SPV FLT n1 n2 n3 (Supervisor fault)
- INT FLT n1 n2 n3 (Interpreter fault)
- P_M FLT n1 n2 n3 (Print Manager fault)
- ENG FLT n1 n2 n3 (Engine Device fault)

2.11 Printer Self-Test

Introduction

Self-test is an internal printer test that checks the printer's logic and electromechanical performance at the user level. A test pattern will be produced on a sheet of printer paper.

Preparation

Before beginning the self-test, verify that:

- the paper and ribbon cartridge are installed correctly.
- the power cable is connected and the printer's power switch is in the off (0) position.
- The printer's top cover is closed.

Self-test procedure

Use the procedure in Table 2-6 to initiate and execute the printer's self-test. Figure 2-3 is an illustration of the printout.

Table 2-6 Signum 2043 Self-Test Procedure

Step	Action
1	Press and hold the ON LINE key while placing the printer's power switch in the on (1) position. Release the ON LINE key. The display shows TEST, and a pattern is printed on the paper as illustrated in Figure 2-3. The printer continues to print until the ON LINE key is pressed a second time. Self-test will end if the printer runs out of paper.
2	The display shows WAIT. You can run self-test with the interface cable disconnected. If the interface cable is not connected to the EPDG I/O or EPDGC I/O board, the printer will display OFFD rather than WAIT when stopped.

Character inspection

Inspect the printout for the following qualities.

- Equal character density
- Character underline and descenders
- Equal character darkness
- Character smudges or strikes
- Character sharpness
- Equal line density
- Character completeness
- Character composition (graphics)

Self-test pattern

Figure 2-4 illustrates the Signum 2043 printer's self-test pattern.

```

ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
BCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
CDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
DEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
EFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
FGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
GHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
HIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
IJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
JKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
LMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
MNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
NOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
OPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
PQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
QRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
RSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
STUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
TUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
UVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
VWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
WXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£
XYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~ ¡¢£

```

3659

Figure 2-4 Signum 2043 Printer Self-Test Print Pattern

2.12 Hex Dump Test

Introduction

The hex dump test should be performed when a protocol line error is detected, and other internal tests do not detect an error. The hex dump test provides a printout of all the characters that were previously transmitted to the printer by the host hardware. Control codes are also included such as ESC, DC4, CR, LF, etc.

Procedure overview

The procedure is executed by entering PROGRAM SETUP, selecting HEX-DUMP PRNT, and choosing the value YES. After initiating the procedure, any character sent to the printer will be printed as its hexadecimal value.

Example: Text in normal mode

ASCII: ESC C H

HEX: 1B 43 48

BASIC: CHR\$(27);CHR\$(67);CHR\$(72)

Example: Same text in hex-dump mode

41 53 43 49 49 3A 20 45 53 43 20 43 20 48 0D 0A 48 45 58 20 20 3A 20 31

42 20 34 33 20 34 38 0D 0A 42 41 53 49 43 3A 20 43 48 52 24 28 32 37 29

3B 43 48 52 24 28 36 37 29 3B 43 48 52 24 28 37 32 29 0D 0A

2.13 Fault Probability Guide

Introduction

Sometimes a fault is detected, but there is no error display.

Usually, if a fault is associated with an electromechanical assembly, no error is detected by the printer firmware because there is no electronic feedback for the fault condition. Specifically, except for the home position and ribbon sensors, the other mechanical parts do not have a return feedback because they are part of an open loop circuit. The home position and ribbon sensors have a feedback circuit to return the status of the part.

Example: electromechanical fault

An example of an electromechanical fault is the paper not advancing upon command. The paper motor, driver circuit, and logic circuitry are involved, but there is no way to identify the part that actually failed. For this fault, only paper movement verification is possible during the T&D printout.

Inconsistent character printing

Table 2-7 is a list of causes and solutions for inconsistent character printing. The character printing may be dark, light, have gaps or spaces, or be incomplete.

Table 2-7 Inconsistent Character Printing

Cause	Solution
Ribbon cartridge installed improperly.	Reinstall the cartridge.
Ribbon cartridge may be faulty.	Install a new cartridge.
Ribbon does not perform correctly (RIBBON BLOCKED error message).	Verify that the carriage unit movement does not interfere with the ribbon movement. If it does, replace the left support group pivot assembly.
Print head cable is installed improperly.	Check the mounting of the print head cable. Reinstall or replace the print head cable.
Print head may be faulty.	Replace the print head.
Main electronics board may be faulty.	Replace the main electronics board.
Power supply electronics board may be faulty.	Replace the power supply electronics board.
Gap between the print head and platen is incorrect.	Adjust the gap between the platen and print head.

Printer stops printing or prints slowly

Table 2-8 lists causes and solutions when the printer stops printing or prints slowly.

Table 2-8 Printing Stops or Slow Printing

Cause	Solution
Thermosensor in the print head is damaged.	It may be necessary to replace the print head if it is not to specifications.
Connection between the print head and the main electronics board is faulty.	Check the connection. It may be necessary to replace the cable.
Main electronics board may be faulty.	It may be necessary to replace the main electronics board.

Color printing quality is poor

Table 2-9 is a list of causes and solutions when the quality of the color printing is poor.

Table 2-9 Color Printing Poor

Cause	Solution
Color ribbon cartridge is installed incorrectly.	Reinstall the color ribbon cartridge.
Color motor is not mounted correctly or is faulty.	Check the mounting of the motor. It may be necessary to replace the motor.
Connection between the print head and the main electronics board is faulty.	Check the connections. It may be necessary to replace the cable.
Main electronics board may be faulty.	It may be necessary to replace the main electronics board.

Character printing misaligned vertically

Table 2-10 is a list of causes and solutions when the character printing is misaligned vertically.

Table 2-10 Character Printing Misaligned Vertically

Cause	Solution
Printer is misaligned.	Run T&D to check vertical printer alignment.
Carriage belt is not adjusted correctly.	The carriage belt may require adjustment.

Tractor/friction solenoid not operable

Table 2-11 is a list of causes and solutions when the tractor/friction solenoid is not operable.

Table 2-11 Tractor/Friction Solenoid Not Operable

Cause	Solution
Tractor/friction solenoid may be faulty.	It may be necessary to replace the tractor/friction solenoid.
Connection between the tractor/friction solenoid and the main electronics board may be faulty.	Check the connection. It may be necessary to replace the cable.
Main electronics board may be faulty.	It may be necessary to replace the main electronics board.

Paper not advancing correctly

Table 2-12 is a list of causes and solutions when the paper does not advance correctly.

Table 2-12 Paper Not Advancing Correctly

Cause	Solution
Connection between the paper motor and the main electronics board is faulty.	Check the connection on the board. It may be necessary to replace the motor.
Paper motor may be faulty.	It may be necessary to replace the motor.
Paper movement does not provide equal line spacing.	The belt tension may require adjustment.
Main electronics board may be faulty.	It may be necessary to replace the main electronics board.
Holes of the paper fanfold are torn.	Remove the torn paper and reload the paper. Adjust the tension of the paper between the left and right tractors.
Paper is not aligned correctly on the sprocket wheels.	Reload the paper and ensure that the corresponding holes on each side of the paper are properly aligned on the tractors.

Printer initialized, display and LEDs blank

Table 2-13 is a list of causes and solutions when the printer initialized, but the display and LEDs are blank. The Operator keys are available.

Table 2-13 Display and LEDs Blank

Solution	Cause
Check the connections. It may be necessary to replace the Operator Panel or main electronics board.	The connection between the Operator Panel and the main electronics board is faulty.

Printer initialized, display blank

Table 2-14 is a list of causes and solutions when the printer initialized, but the display is blank. The Operator keys are not available.

Table 2-14 Display Blank

Cause	Solution
The connection between the Operator Panel and the main electronics board is faulty.	Check the connections. It may be necessary to replace the Operator Panel or main electronics board.
The ribbon sensor is faulty.	Check the ribbon sensor action through the pivot assembly. It may be necessary to replace the pivot assembly, the sensors cable, or the ribbon motor assembly.

Printer initialized, carriage assembly moves 1 inch to right

Table 2-15 is a list of causes and solutions when the printer initialized, but the carriage assembly moved 1 inch to the right.

Table 2-15 Carriage Assembly Moves One Inch Right

Cause	Solution
The connection to the home position sensor or the carriage motor is faulty.	Check the home position sensor connection and carriage motor connection on the main electronics board.
The home position sensor or carriage motor assembly are faulty.	Check the ribbon sensor action through the carriage assembly and the impedance of the motor. It may be necessary to replace the home position sensor cable or the carriage motor assembly.

Printer not initialized, buzzer sounds continuously

Table 2-16 is a list of causes and solutions when the printer did not initialize and the buzzer sounds continuously.

Table 2-16 Printer Not Initialized and Buzzer Sounds

Cause	Solution
The Controller EPROM or the main electronics board may be faulty.	It may be necessary to replace the Controller EPROM or main electronics board.

2.14 Printer Replacement

CAUTION

When replacing a printer, be sure that the power source line voltage is the same as the voltage requirement displayed at the back of the printer. If the line voltage does not match the printer voltage, do not use the printer; instead, consult your Honeywell representative.

Introduction

The printer is the Optimal Replaceable Unit (ORU). A few lower level replacement parts are also currently (1996) available by factory order.

Removal procedure

Use the procedure in Table 2-17 to disconnect and remove a printer.

Table 2-17 Printer Removal Procedure

Step	Action
1	Place the printer's power switch that is located at the left side of the printer in the off (0) position.
2	Disconnect the ac power cord.
3	Disconnect the data cable from its connector at rear of the printer.
4	Remove the printer.

Installation procedure

Use the procedure in Table 2-18 to install a printer.

Table 2-18 Printer Installation Procedure

Step	Action
1	Lift the printer from its shipping box by firmly grasping both sides of the printer. Temporarily place the printer on a flat work surface.
2	Remove the printer accessories that are included in the box. The accessories should include the following items. • ribbon cartridge • tractor unit • power cable • driver diskette An optional second tractor unit and color ribbon cartridge in separate boxes may be included with the printer.
3	If a ribbon cartridge is not installed in the printer, use the procedure in subsection 2.16 to install a ribbon cartridge.
4	If the printer must operate with one or two tractors, use the procedure in subsection 2.17 to install the tractor(s).

Table 2-18 Printer Installation Procedure

Step	Action
5	Choose a suitable location for the printer. Consider the following. • The distance between the printer and the US or U^XS module must not exceed the length of the interface cable. • The location must be sturdy, horizontal, and stable. • The printer must not be exposed to direct sunlight, extreme heat, cold, dust, or high humidity. • The ac power outlet must be near the printer and compatible with the printer's power cord plug. Ground must be provided through the plug. The voltage at the outlet must match the voltage shown on the printer's name plate. Place the printer in its operating location.
6	Connect one end of the 25-pin interface cable to the I/O interface board (EPDG I/O or EPDGC I/O) in the US or U^XS module that will control the printer and the other end of the cable to the printer. To secure the connections, tighten the connector screws at both ends of the cable.
7	Verify that the printer's power switch that is located at the left side of the printer is in the off (0) position. Connect the power cord to the power connector at the rear of the printer.
8	Place the power switch in the on (1) position. The print head should move to the left side of the printer and stop.

2.15 Ribbon Cartridge Replacement

CAUTION

The print head can become hot during extended operation. Be sure it is cool enough to touch before starting to remove the printer cartridge.

Printer cartridge replacement procedure

Use the procedure in Table 2-19 to replace a printer ribbon cartridge

Table 2-19 Ribbon Cartridge Replacement Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	Open the printer cover by pressing on the release buttons.
3	Move the printer carriage to the center of the printer.
4	Pull the paper thickness lever towards the front of the printer to facilitate the removal of the ribbon cartridge.
5	Grasp the ribbon cartridge firmly at its sides and lift it straight up until the side clips release it.
6	Remove the new ribbon cartridge from its bag. Remove and discard the hold fast that blocks the ribbon.
7	Before installing the ribbon cartridge, turn the ribbon winding knob on the cartridge in the direction of the arrow on the cartridge to take up any ribbon slack.
8	Insert the ribbon between the print head and the print head mask. Lay the cartridge over the printer carriage. Make sure the two pins on each side of the cartridge are positioned over the white retaining clips of the printer carriage.
9	Push the cartridge gently down while turning the tension knob. Make sure that the cartridge clips into place.
10	To tighten the ribbon, turn the tension knob in the direction of the arrow on the ribbon cartridge.
11	To ensure that the ribbon guide is free, manually move the print carriage from right to left and back.
12	Move the paper thickness lever towards the back of the printer and adjust it to

Table 2-19 Ribbon Cartridge Replacement Procedure

Step	Action
	obtain the best quality printing.
13	Check the quality of print with the self-test procedure in subsection 2.11. If the pattern is not satisfactory, adjust the paper thickness lever accordingly.
14	Close the front printer cover.
15	Place the printer's power switch in the on (1) position.

2.16 Tractor Installation/Removal

Introduction

A paper tractor can be installed in three positions. One position is in the front of the printer, and two positions are at the rear of the printer.

The front position provides push paper feeding, and the rear positions are for either push or pull paper feeding.

CAUTION

Be sure the printer's power switch is in the off (0) position before installing or removing a tractor.

Front position installation/removal

Use the procedure in Table 2-20 to install or remove a tractor in the front position.

Table 2-20 Tractor Front Position Installation/Removal Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	To install the tractor, lift the front tractor cover and fasten it to the single sheet support cover.
3	Lift the tractor connector protector at the right side of the printer. Insert the tractor with its connector on the right by pushing it upward until it clicks.
4	Close the front tractor cover.
5	To remove the tractor, press the two small levers at both sides of the tractor that secure the tractor in place and pull the tractor downward. Close the tractor connector protector.
6	Close the front tractor cover.
7	Place the printer's power switch in the on (1) position.

Front tractor push operation

Figure 2-5 illustrates front tractor push operation.

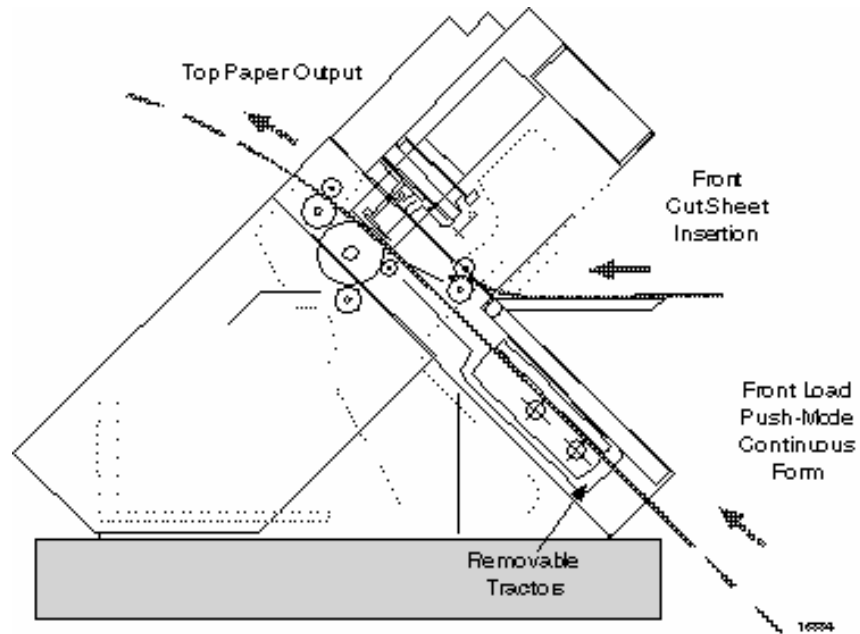


Figure 2-5 Front Tractor Push Operation

Rear push position installation/removal

Use the procedure in Table 2-21 to install or remove a tractor in the rear push position.

Table 2-21 Tractor Rear Push Position Installation/Removal Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	Remove the rear tractor cover.
3	Set the slider on the tractor in the push position (down position).
4	Orient the tractor with its grooves facing downward and its connector on the left side.
5	Insert the left and right tractor grooves into the printer pins until the tractor clicks in place.
6	To remove the tractor, press the two small levers that secure the tractor in place and pull upward.
7	Replace the rear tractor cover.
8	Place the printer's power switch in the on (1) position.

Rear tractor push operation

Figure 2-6 illustrates rear tractor push operation.

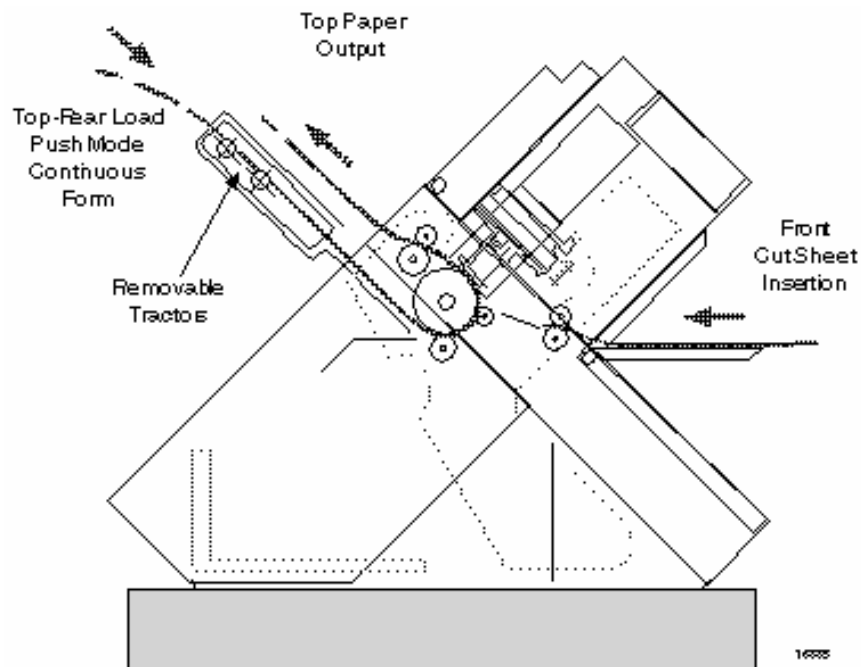


Figure 2-6 Rear Tractor Push Operation

Rear pull position installation/removal

Use the procedure in Table 2-22 to install or remove a tractor in the rear pull position.

Table 2-22 Tractor Rear Pull Position Installation/Removal Procedure

Step	Action
1	Place the printer's power switch in the off (0) position.
2	Remove the rear tractor cover.
3	Set the slider on the tractor in the pull position (up position).
4	Move the paper bail to the forward position.
5	Orient the tractor with its grooves facing downward and its connector on the left side.
6	Insert the left and right tractor grooves into the printer pins until the tractor clicks in place.
7	Position the levers under the tractor down to unlock them and gently press the tractor downward until it clicks in place.
8	To remove the tractor, lift the tractor, press the two small levers that secure the tractor, and pull upward.
9	Replace the rear tractor cover.
10	Place the printer's power switch in the on (1) position.

Rear tractor pull operation

Figure 2-7 illustrates rear tractor pull operation.

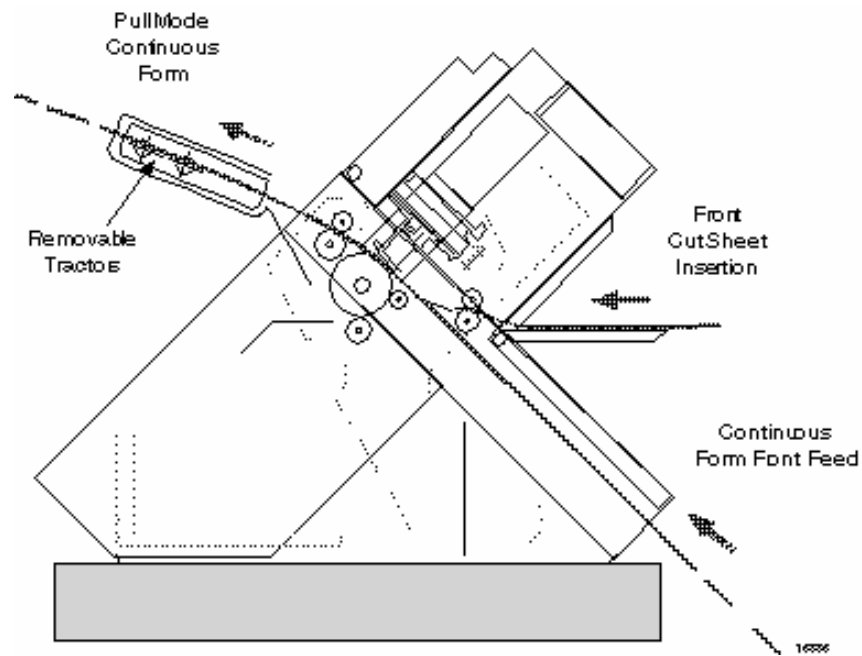


Figure 2-7 Rear Tractor Pull Operation

Dual tractor operation

Dual tractors can be installed in both the front and rear positions simultaneously. Continuous paper can be loaded at either the front or rear of the printer.

Front tractor push or rear tractor pull operation

Figure 2-8 illustrates either front tractor push operation or rear tractor pull operation.

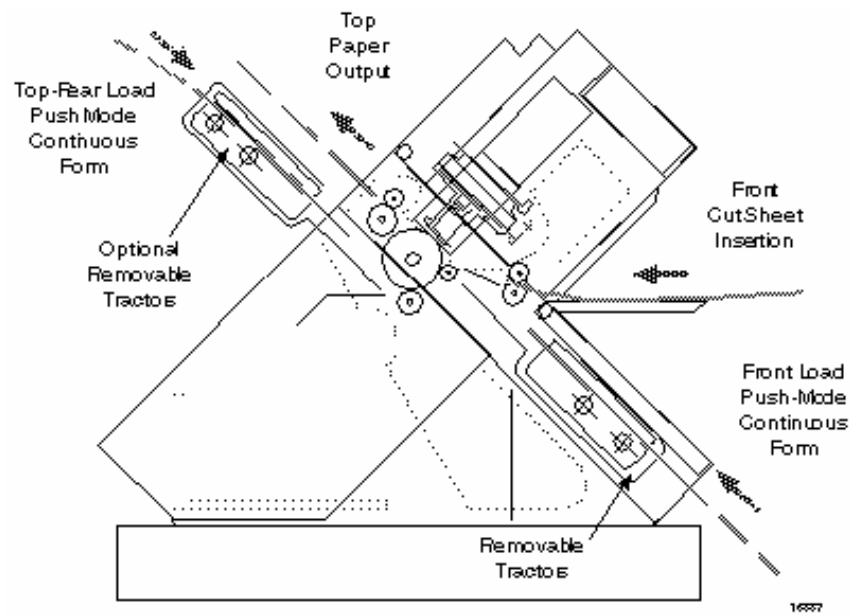


Figure 2-8 Front Tractor Push or Rear Tractor Pull Operation

Front tractor push and rear tractor pull operation

Figure 2-9 illustrates front tractor push operation and rear tractor pull operation.

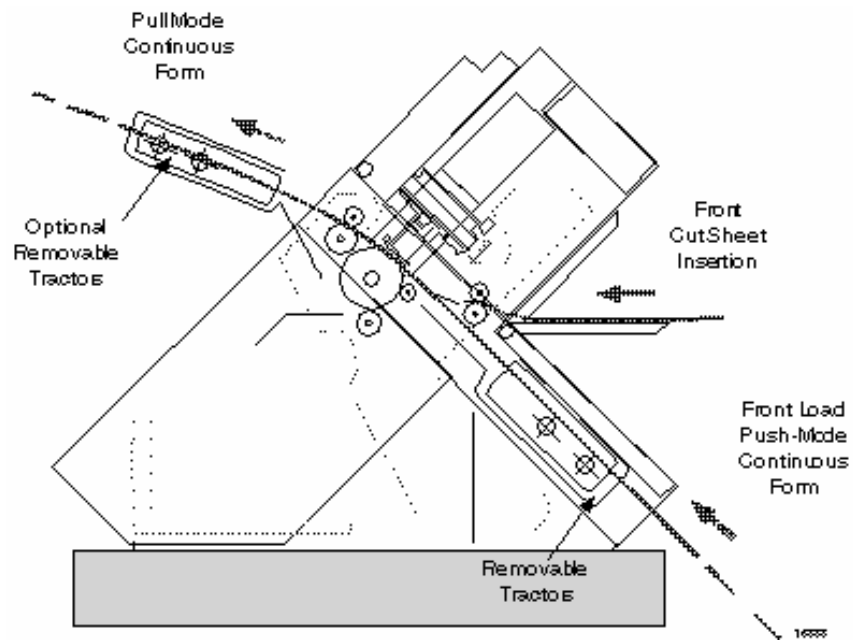


Figure 2-9 Front Tractor Push and Rear Tractor Pull Operation

2.17 Print Head Replacement

Replacement procedure

Use the procedure in Table 2-23 to remove and replace a print head.

Table 2-23 Print Head Removal/Replacement Procedure

Step	Action
1	Place the printer's power switch in the off (0) position. Be sure the print head is not too hot to touch.
2	Depress the pushbuttons on each side of the top cover and open the cover.
3	Use the procedure in subsection 2.15 to remove the ribbon cartridge.
4	Loosen and remove the screw that secures the print head cable to the carriage assembly.
5	Disconnect the print head cable from the print head.
6	Loosen and remove the two screws that secure the print head to the carriage assembly.
7	Remove the print head from the carriage assembly.
8	Install a new print head by performing the above steps in reverse order.
9	Run the self-test (subsection 2.11) to verify that the new print head is properly installed.

Print head replacement

Figure 2-10 illustrates replacement of the print head.

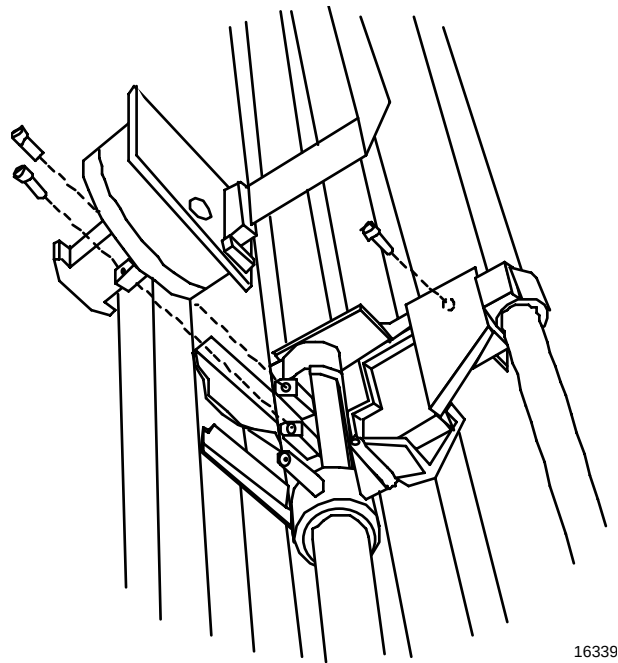


Figure 2-10 Print Head Replacement

2.18 Spare Parts

Optimal Replaceable Units (ORUs)

Table 2-24 is a list of Optimal Replaceable Units (ORU) for the Signum 2043 printer that can be purchased from Honeywell. Other than a replacement print head and a tractor, no other electromechanical assemblies or electronic board assemblies are considered ORU parts.

Table 2-24 Optimal Replaceable Units (ORU) Parts List

Description	Part Number
Signum 2043 Printer, 120 Vac	51196690-100
Signum 2043 Printer, 240 Vac	51196690-200
Ribbon Cartridge, Black	51196690-400
Ribbon Cartridge, Color	51196690-501
Paper Tractor Unit	51196690-800
Printer Floor Pedestal	51196690-900
Print Head	51196690-920

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